CHAPTER 34

Pharynx

The pharynx is a 12–14 cm long musculomembranous tube shaped like an inverted cone. It extends from the cranial base to the lower border of the cricoid cartilage (the level of the sixth or seventh cervical vertebra), where it becomes continuous with the oesophagus. The width of the pharynx varies constantly because it is dependent on muscle tone, especially of the constrictors; at rest, the pharyngo-oesophageal junction is closed as a result of tonic closure of the upper oesophageal sphincter, and during sleep, muscle tone is low and the dimensions of the pharynx are markedly decreased (which may give rise to snoring and sleep apnoea). The pharynx is limited above by the posterior part of the body of the sphenoid and the basilar part of the occipital bone, and it is continuous with the oesophagus below. Behind, it is separated from the cervical part of the vertebral column and the prevertebral fascia, which covers longus colli and longus capitis, by loose connective tissue in the retropharyngeal space above and the retrovisceral space below.

The muscles of the pharynx are three circular constrictors and three longitudinal elevators. The constrictors may be thought of as three overlapping cones that arise from structures at the sides of the head and neck, and pass posteriorly to insert into a midline fibrous band, the pharyngeal raphe. The arterial supply to the pharynx is derived from branches of the external carotid artery, particularly the ascending pharyngeal artery, but also from the ascending palatine and tonsillar branches of the facial artery, the maxillary artery (greater palatine and pharyngeal arteries and the artery of the pterygoid canal) and dorsal lingual branches of the lingual artery. The pharyngeal veins begin in a plexus external to the pharynx, receive meningeal veins and a vein from the pterygoid canal, and usually end in the internal jugular vein. Lymphatic vessels from the pharynx and cervical oesophagus pass to the deep cervical nodes, either directly or through the retropharyngeal or paratracheal nodes. The motor and sensory innervation is principally via branches of the pharyngeal plexus.

The pharynx lies behind, and communicates with, the nasal, oral and laryngeal cavities via the nasopharynx, oropharynx and laryngopharynx, respectively (**Figs 34.1–34.2**). Its lining mucosa is continuous with that lining the pharyngotympanic tubes, nasal cavity, mouth and larynx. The retropharyngeal and parapharyngeal spaces surround the pharynx; the retropharyngeal space lies anterior to the prevertebral and alar fascia and thus to the alar, danger space that lies between them. For further reading, including reviews of some of the more important historical literature, see Flint et al (2010), Graney et al (1998), Hollinshead (1982), Wood-Jones (1940).

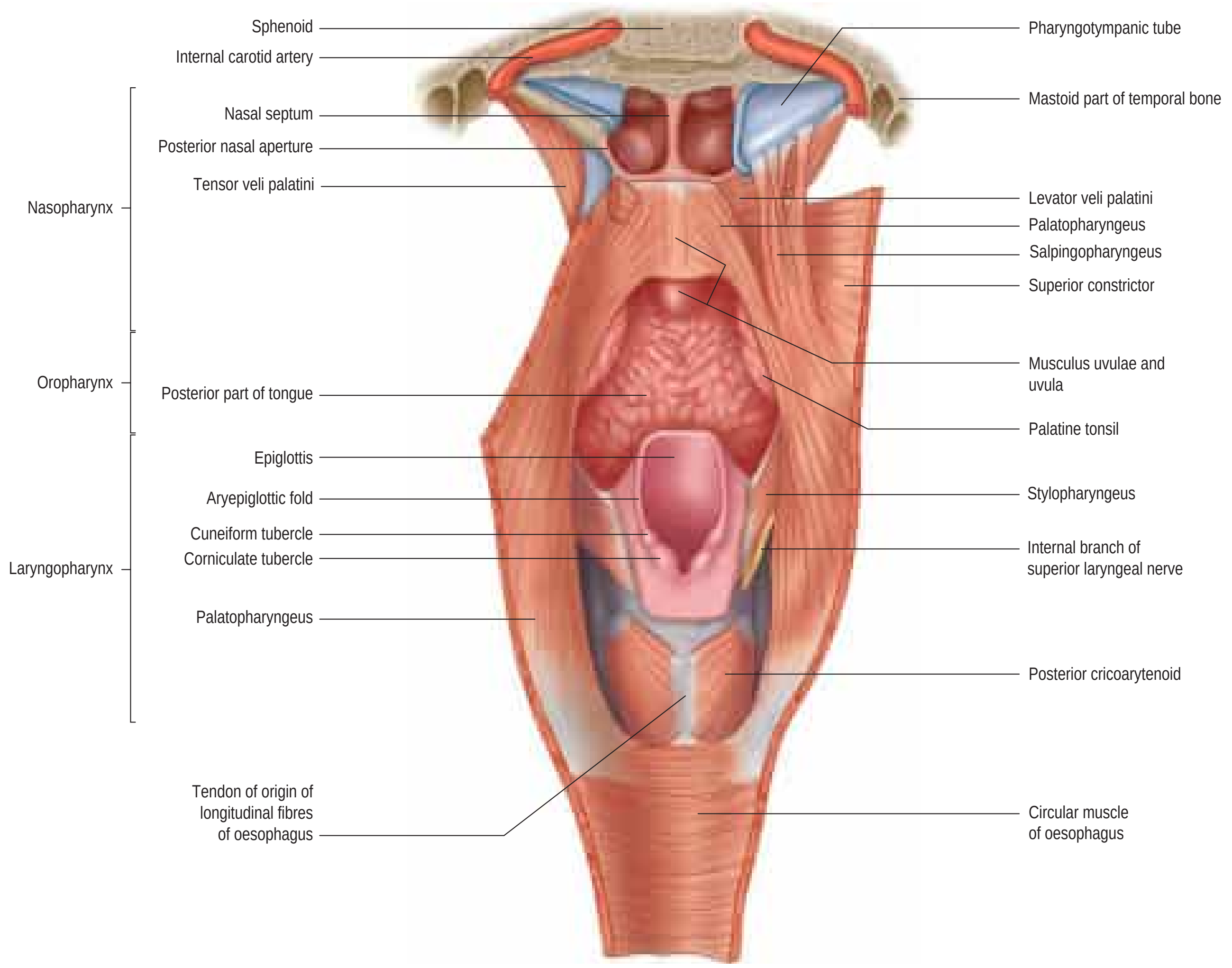

Fig. 34.1 The nasopharynx, oropharynx and laryngopharynx, exposed by cutting the median pharyngeal raphe and reflecting the constrictor muscles laterally on either side, posterior view.

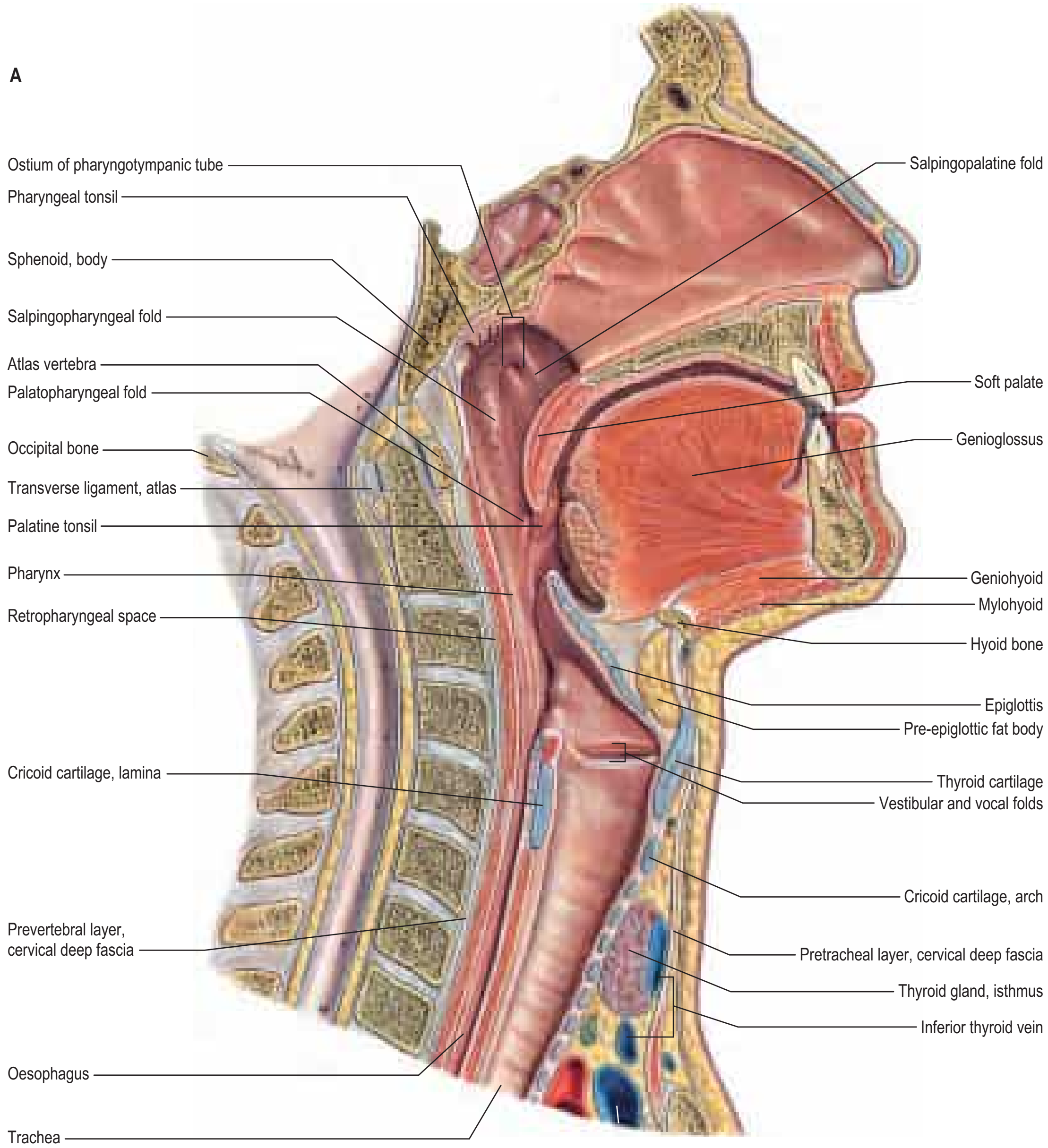

Fig. 34.2 **A**, A sagittal section through the head and neck, including the nasal and oral cavities but excluding the intracranial region. **B**, A corresponding magnetic resonance image (MRI), which also includes the posterior cranial fossa, cerebellum and cervical spinal cord. Key: 1, hard palate; 2, soft palate; 3, uvula; 4, epiglottis; 5, hyoid bone; 6, dens of axis; 7, lamina of cricoid cartilage; 8, oesophagus. (A, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer, 2013.)

NASOPHARYNX

Boundaries

The nasopharynx lies above the soft palate and behind the posterior nares, which allow free respiratory passage between the nasal cavities and the nasopharynx (see Figs 34.1–34.2). The nasal septum separates the two posterior nares, each of which measures approximately 25 mm vertically and 12 mm transversely. Just within these openings lie the posterior ends of the inferior and middle nasal conchae/turbinates (Ch. 33). The nasopharynx has a roof, a posterior wall, two lateral walls and a floor. These are rigid (except for the floor, which can be raised or lowered by the soft palate), and the cavity of the nasopharynx is therefore never obliterated by muscle action, unlike the cavities of the oro- and laryngopharynx. The nasal and oral parts of the pharynx

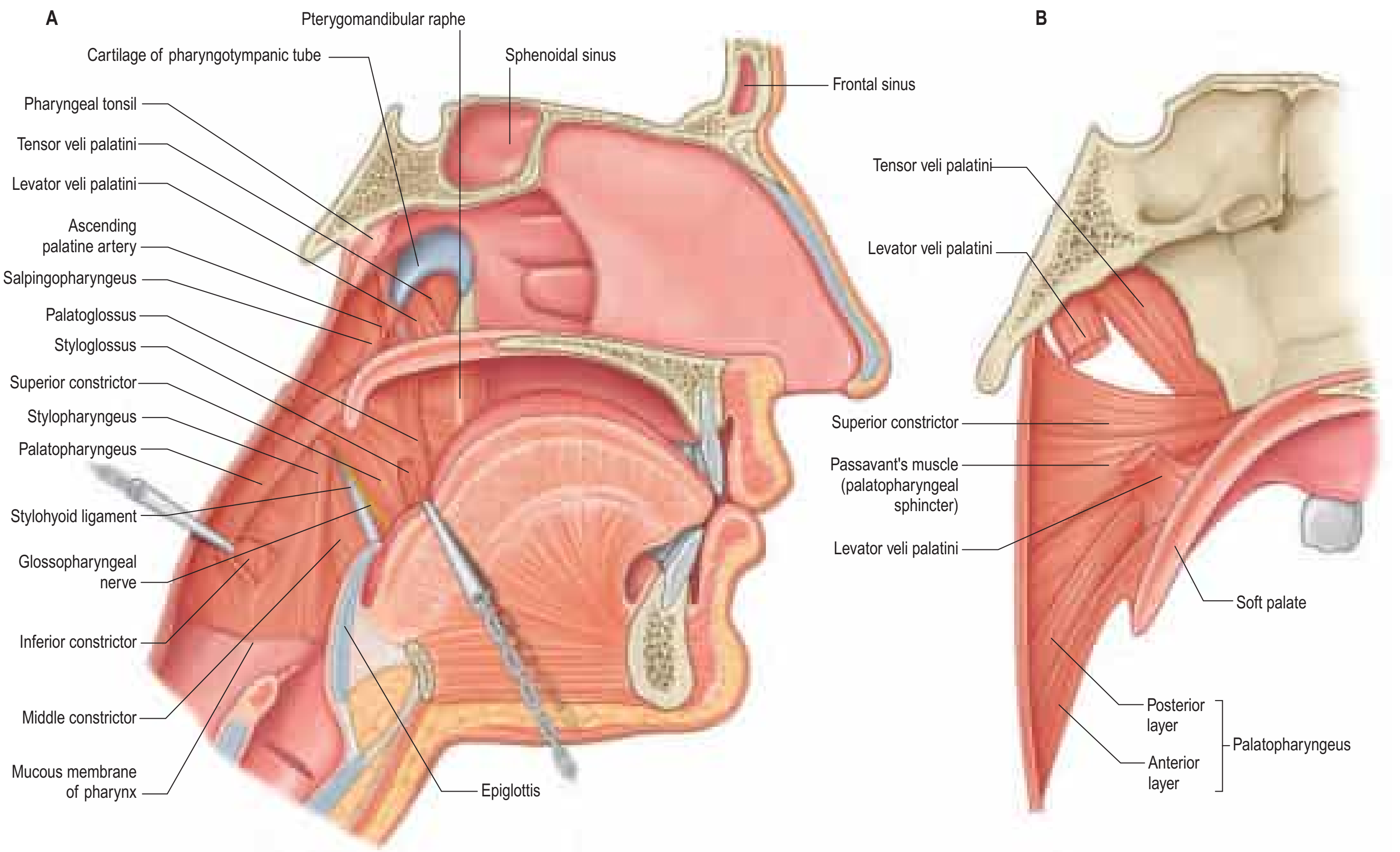

Fig. 34.3 **A**, The interior of the pharynx, exposed by removal of the mucous membrane, sagittal section. The bodies of the cervical vertebrae have been removed, the cut posterior wall of the pharynx retracted dorsolaterally and palatopharyngeus reflected dorsally to show the cranial fibres of the inferior constrictor. The dorsum of the tongue has been pulled ventrally to display a part of styloglossus in the angular interval between the mandibular and the lingual fibres of origin of the superior constrictor. **B**, Muscles of the left half of the soft palate and adjoining part of the pharyngeal wall, sagittal section.

communicate through the pharyngeal isthmus, which lies between the posterior border of the soft palate and the posterior pharyngeal wall. Elevation of the soft palate and constriction of the palatopharyngeal sphincter close the isthmus during swallowing.

The roof and posterior wall form a continuous concave slope that leads down from the nasal septum to the oropharynx. It is bounded above by mucosa overlying the posterior part of the body of the sphenoid, and further back by the basilar part of the occipital bone as far as the pharyngeal tubercle. Further down, the mucosa overlies the pharyngobasilar fascia and the upper fibres of the superior constrictor, and behind these, the anterior arch of the atlas. A lymphoid mass, the pharyngeal tonsil (adenoid, nasopharyngeal tonsil), lies in the mucosa of the upper part of the roof and posterior wall in the midline.

The lateral walls of the nasopharynx display a number of important surface features. On either side, each receives the opening of the pharyngotympanic tube (auditory or Eustachian tube), situated 10–12 mm behind and a little below the level of the posterior end of the inferior nasal concha (see Fig. 34.2). The tubal aperture is approximately triangular in shape, and is bounded above and behind by the tubal elevation that consists of mucosa overlying the protruding pharyngeal end of the cartilage of the pharyngotympanic tube. A vertical mucosal fold, the salpingopharyngeal fold, descends from the tubal elevation behind the aperture (see Fig. 34.2) and covers salpingopharyngeus in the wall of the pharynx (**Fig. 34.3**); a smaller salpingopalatine fold extends from the anterosuperior angle of the tubal elevation to the soft palate in front of the aperture. As levator veli palatini enters the soft palate, it produces an elevation of the mucosa immediately around the tubal opening (see Fig. 34.3). A small variable mass of lymphoid tissue, the tubal tonsil, lies in the mucosa immediately behind the opening of the pharyngotympanic tube. Further behind the tubal elevation is a variable depression in the lateral wall, the lateral pharyngeal recess (fossa of Rosenmüller), located between the posterior wall of the nasopharynx and the salpingopharyngeal fold. The floor of the nasopharynx is formed by the nasal, upper surface of the soft palate.

The relations of the nasopharynx are important in understanding the spread of nasopharyngeal carcinoma (Chong and Ong 2008). The mucosa of the nasopharynx is separated from the masticator space by the parapharyngeal space. The carotid sheath, containing the carotid space, lies posterior and lateral to the parapharyngeal space. The glossopharyngeal, vagus, accessory and hypoglossal nerves lie within the upper part of the carotid sheath and they come to lie within the superior part of the parapharyngeal space, along with the sympathetic chain. The foramen lacerum lies superolateral to the fossa of Rosenmüller; in life, the foramen is closed by cartilage, over which the internal carotid artery runs. The foramen ovale, which transmits the mandibular division of the trigeminal nerve, lies still further laterally; the mandibular nerve passes through the parapharyngeal space and then the masticator space in order to innervate the muscles of mastication. The retropharyngeal space is posterior to the nasopharynx.

Microstructure of non-tonsillar nasopharynx The nasopharyngeal epithelium anteriorly is ciliated, pseudostratified respiratory in type with goblet cells (see Fig. 2.2D). The ducts of mucosal and submucosal seromucous glands open on to its surface. Posteriorly, the respiratory epithelium changes to non-keratinized stratified squamous epithelium which continues into the oropharynx and laryngopharynx. The transitional zone between the two types of epithelium consists of columnar epithelium with short microvilli instead of cilia. Superiorly, this zone meets the nasal septum; laterally, it crosses the orifice of the pharyngotympanic tube; and it passes posteriorly at the union of the soft palate and the lateral wall. Numerous mucous glands surround the tubal orifices.

Innervation Much of the mucosa of the nasopharynx behind the pharyngotympanic tube is supplied by the pharyngeal branch of the pterygopalatine ganglion, which traverses the palatovaginal canal with a pharyngeal branch of the maxillary artery.

PHARYNGEAL TONSIL

The pharyngeal tonsil (adenoid, nasopharyngeal tonsil) is a median mass of mucosa-associated lymphoid tissue (MALT; see Ch. 4) situated in the roof and posterior wall of the nasopharynx (**Fig. 34.4**). At its maximal size (during the early years of life), it is shaped like a truncated pyramid, often with a vertically orientated median cleft, so that its apex points towards the nasal septum and its base to the junction of the roof and posterior wall of the nasopharynx (Papaioannou et al 2013).

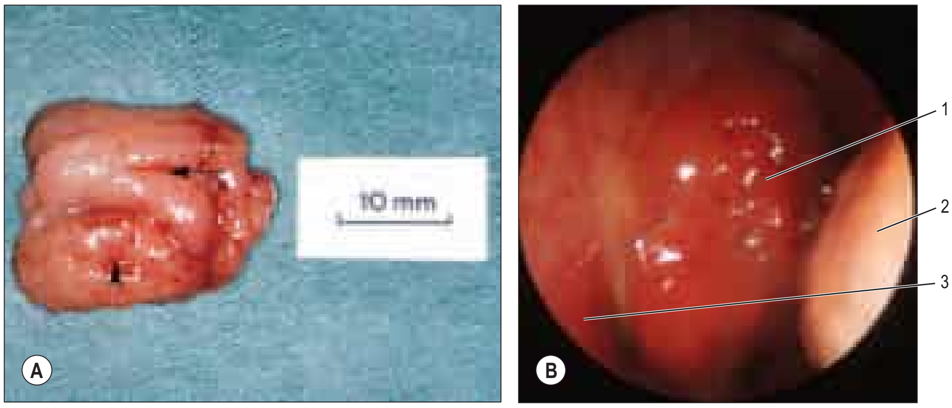

Fig. 34.4 **A**, A pharyngeal tonsil following adenoidectomy by curettage. The rostral surface is to the left; surface folds radiate forwards from a median recess (arrow). In this example, the impression left by contact with the left Eustachian cushion is evident laterally (arrowhead). **B**, A transnasal endoscopic view of a pharyngeal tonsil. Key: 1, pharyngeal tonsil (in posterior naris); 2, inferior concha (posterior view); 3, posterior end of nasal septum. (A, Specimen provided by Professor MJ Gleeson, School of Medicine, King's College London. B, Courtesy of Mr Simon A Hickey.)

The free surface of the pharyngeal tonsil is marked by folds that radiate forwards and laterally from a median blind recess, the pharyngeal bursa (bursa of Luschka), which extends backwards and up. The recess is present in the fetus and the young but only occasionally present in the adult, and marks the rostral end of the embryological notochord. The number and position of the folds and of the deep fissures that separate them vary. A median fold may pass forwards from the pharyngeal bursa towards the nasal septum, or instead a fissure may extend forwards from the bursa, dividing the nasopharyngeal tonsil into two distinct halves that reflect its paired developmental origins (see Fig. 34.4).

The prenatal origins of the pharyngeal tonsil are described on page 615. After birth, it initially grows rapidly, but usually undergoes a degree of involution and atrophy from the age of 8–10 years (although hypoplasia may still occur in adults up to the seventh decade). Relative to the volume of the nasopharynx, the size of the tonsil is largest at 5 years, which may account for the frequency of nasal breathing problems in preschool children, and the incidence of adenoidectomy in this age group.

Vascular supply and lymphatic drainage The arterial supply of the pharyngeal tonsil is derived from the ascending pharyngeal and ascending palatine arteries, the tonsillar branches of the facial artery, the pharyngeal branch of the maxillary artery and the artery of the pterygoid canal (see Figs 34.6–34.7). In addition, a nutrient or emissary vessel to the neighbouring bone, the basisphenoidal artery, which is a branch of the inferior hypophysial arteries, supplies the bed of the pharyngeal tonsil and is a possible cause of persistent post-adenoidectomy haemorrhage in some patients.

Numerous communicating veins drain the pharyngeal tonsil into the internal submucous and external pharyngeal venous plexuses. They emerge from the deep lateral surface of the tonsil and join the external palatine (paratonsillar) veins, and pierce the superior constrictor either to join the pharyngeal venous plexus, or to unite to form a single vessel that enters the facial or internal jugular vein; they may also connect with the pterygoid venous plexus.

Microstructure The pharyngeal tonsil is covered laterally and inferiorly mainly by ciliated respiratory epithelium that contains scattered small patches of non-keratinized stratified squamous epithelium. Its superior surface is separated from the periosteum of the sphenoid and occipital bones by a connective tissue hemicapsule. The fibrous framework of the tonsil, consisting of a mesh of collagen type III (reticular) fibres supporting a lymphoid parenchyma similar to that in the palatine tonsil, is anchored to the hemicapsule.

The nasopharyngeal epithelium lines a series of mucosal folds around which the lymphoid parenchyma is organized into follicles and extrafollicular areas. Internally, the tonsil is subdivided into 4–6 lobes by connective tissue septa, which arise from the hemicapsule and penetrate the lymphoid parenchyma. Seromucous glands lie within the connective tissue, and their ducts extend through the parenchyma to reach the nasopharyngeal surface.

Functions The pharyngeal tonsil forms part of the circumpharyngeal lymphoid ring (Waldeyer's ring) and therefore presumably contributes to the defence of the upper respiratory tract. The territories served by its lymphocytes are uncertain but may include the nasal cavities, nasopharynx, pharyngotympanic tubes and the middle and inner ears.

Adenoidectomy Surgical removal of the pharyngeal tonsil (adenoid) is commonly performed to clear nasopharyngeal obstruction and as part of the treatment of chronic secretory otitis media. A variety of methods are employed, including suction diathermy, suction microdebridement and, most commonly, blind curettage. When using the latter, it is important to avoid hyperextension of the cervical spine, as this throws the arch of the atlas into prominence and may result in damage to the prevertebral fascia and anterior spinal ligaments, with resultant infection and cervical instability. Extreme lateral curettage can result in damage to the tubal orifice and excessive bleeding because the vasculature is denser laterally. Removal of the pharyngeal tonsil in children can result in an impairment in the ability of the soft palate to close the pharyngeal isthmus fully (velopharyngeal insufficiency, VPI), causing excessive nasality of speech (Gray and Pinborough-Zimmerman 1998).

PHARYNGOTYMPANIC TUBE

The pharyngotympanic tube (see Figs 34.9A, B, 37.6–37.7) connects the tympanic cavity to the nasopharynx. It has several functions related to maintaining the health of the middle ear, including pressure equalization on both aspects of the tympanic membrane, mucociliary clearance and 'drainage', and protection from the influences of the nasopharyngeal environment and loud sounds. Approximately 36 mm long, it descends anteromedially from the tympanic cavity to the nasopharynx at an angle of approximately 45° with the sagittal plane and 30° with the horizontal (these angles increase with age and elongation of the cranial base). It is formed partly by cartilage and fibrous tissue, and partly by bone.

The cartilaginous part, which is approximately 24 mm long, is formed by a triangular plate of cartilage, the greater part of which is in the posteromedial wall of the tube. Its apex is attached by fibrous tissue to the circumference of the jagged rim of the bony part of the tube, and its base is directly under the mucosa of the lateral nasopharyngeal wall, forming a tubal elevation (torus tubarius; Eustachian cushion) behind the pharyngeal orifice of the tube (see Fig. 31.1). The upper part of the cartilage is bent laterally and downwards, producing a broad medial lamina and narrow lateral lamina. In transverse section, it is hook-like and incomplete below and laterally, where the canal is composed of fibrous tissue. The cartilage is fixed to the cranial base in the groove between the petrous part of the temporal bone and the greater wing of the sphenoid, and ends near the root of the medial pterygoid plate. In adults, the cartilaginous and bony parts of the tube are not in the same plane, the former descending a little more steeply than the latter. The diameter of the tube is greatest at the pharyngeal orifice and least at the junction of the two parts (the isthmus), increasing again towards the tympanic cavity. At birth, the pharyngotympanic tube is about half its adult length; it is more horizontal and its bony part is relatively shorter but much wider.

The bony part, approximately 12 mm long, is oblong in transverse section, with its greater dimension in the horizontal plane. It starts from the anterior tympanic wall and gradually narrows to end at the junction of the squamous and petrous parts of the temporal bone, where it has a jagged margin for the attachment of the cartilaginous part. The carotid canal lies medially.

The mucosa of the pharyngotympanic tube is continuous with the nasopharyngeal and tympanic mucosae. The bony canal is lined by a prolongation of the low cuboidal, ciliated mucosa that lines the middle ear. The histology changes to pseudostratified, ciliated, columnar epithelium typical of the upper respiratory tract after the bony/cartilaginous junction. Within the cartilaginous tube, the floor contains numerous mucus-producing goblet cells and is heavily rugated, whereas the walls in the upper half of the tube contain fewer goblet cells and are generally smooth. Mucosal folding is a feature of the childhood tube; the increased surface area and concomitant increased numbers of ciliated

Thornwaldt's cyst is an embryonic remnant of the persisting cranial end of the notochord that sometimes develops if the pharyngeal bursa is occluded (Fig. 34.5). Usually benign and asymptomatic, these cysts can become infected; symptoms include halitosis, occipital headache and postnasal drip (Yuca and Varsak 2012).

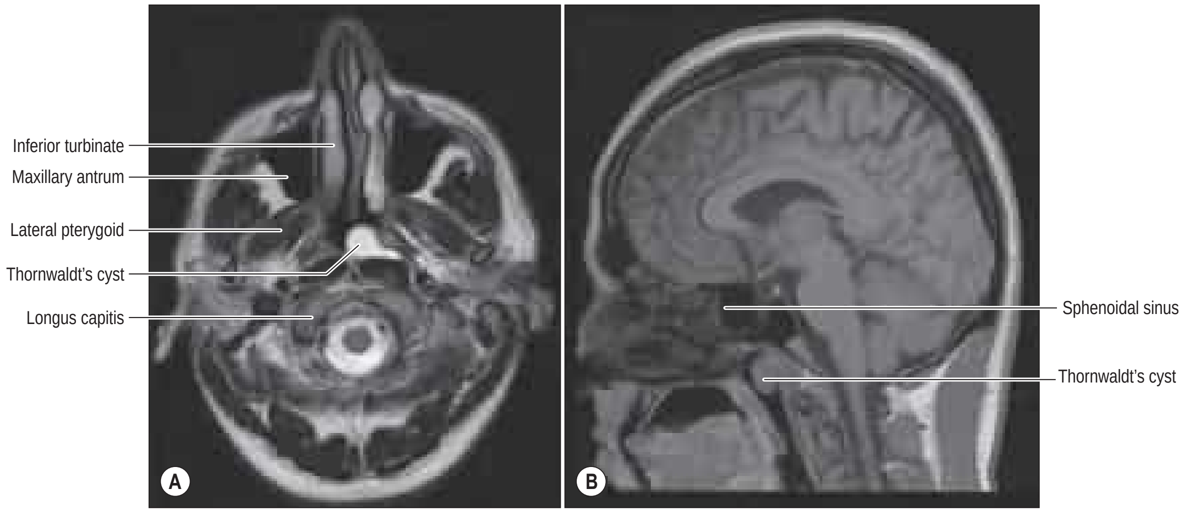

Fig. 34.5 Thornwaldt's cyst. **A**, axial view; **B**, sagittal view.

cells are presumed to facilitate improved clearance of the mucociliary blanket. Folds decrease with age up to the age of 20 years, when adult characteristics are reached. Mucous glands predominate in neonates; mucous, serous and mixed glands are present in equal amounts in young children; and serous glands predominate in later life (Orita et al 2002). The smoothness of the wall in the upper half of the tube may function in pressure equalization and gas exchange. The pharyngeal orifice is a narrow slit, level with the palate and without a tubal elevation. A lymphoid mass, the tubal tonsil, is found near the pharyngeal orifice; it is variable and sometimes considerable. (For further reading see Sade (1989), Bluestone (1998), O'Reilly and Sando (2010).)

Relations Salpingopharyngeus is attached to the inferior part of the cartilaginous tube near its pharyngeal opening. Posteromedially are the petrous part of the temporal bone and levator veli palatini, which arises partly from the medial lamina of the tube. Anterolaterally, tensor veli palatini separates the tube from the otic ganglion, the mandibular nerve and its branches, the chorda tympani nerve and the middle meningeal artery.

Vascular supply The osseous part of the pharyngotympanic tube is supplied by the tubal artery (a branch of the accessory meningeal artery) and the caroticotympanic branches of the internal carotid artery. The cartilaginous part of the tube is supplied by the deep auricular and pharyngeal branches of the maxillary artery, the ascending palatine artery (usually a branch of the facial artery but sometimes branching directly from the external carotid artery) and the ascending pharyngeal branch of the external carotid artery. The veins of the pharyngotympanic tube usually drain to the pterygoid venous plexus.

Innervation The pharyngotympanic tube is innervated by filaments from the tympanic plexus and from the pharyngeal branch of the pterygopalatine ganglion. The tympanic plexus ramifies on the promontory in the middle ear cavity and is formed by the tympanic branch of the glossopharyngeal nerve and caroticotympanic nerves of sympathetic origin (see Fig. 37.13).

Equalization of pressure At rest, the cartilaginous part of the pharyngotympanic tube is closed at the nasopharyngeal orifice. In normal individuals, the tube opens in order to equalize middle ear pressure to ambient pressure. Analysis of slow-motion video-endoscopy reveals four stages in tubal opening. Initially, the soft palate is elevated, the lateral pharyngeal wall moves medially and the medial lamina of the pharyngotympanic tube rotates medially (levator veli palatini is thought to open the distal part of the tube). The lateral wall of the pharyngotympanic tube moves laterally, so that the orifice is dilated both laterally and vertically. Tubal dilation propagates from distal to proximal by the action of dilator tubae. Finally, the proximal cartilaginous tube adjacent to the junctional region opens. Opening usually lasts for 0.3–0.5 sec but is prolonged during yawning. Closure depends on adhesion of the intraluminal mucous blanket, elastic forces of the supporting tissues, and hydrostatic pressure of venous blood.

OROPHARYNX

Boundaries

The oropharynx extends from below the soft palate to the upper border of the epiglottis (see Figs 34.1–34.2). It opens into the mouth through the oropharyngeal isthmus, demarcated by the palatoglossal arch, and faces the pharyngeal aspect of the tongue. Its lateral wall consists of the palatopharyngeal arch and palatine tonsil (see Fig. 31.3). Posteriorly, it is level with the bodies of the second, and upper part of the third, cervical vertebrae (see Fig. 34.2).

SOFT PALATE

The soft palate is a mobile flap suspended from the posterior border of the hard palate, sloping down and back between the oral and nasal parts of the pharynx (see Figs 34.2–34.3). The boundary between the hard and soft palate is readily palpable and may be distinguished by a change in colour, the soft palate being a darker red with a yellowish tint. The soft palate is a thick fold of mucosa enclosing an aponeurosis, muscular tissue, vessels, nerves, lymphoid tissue and mucous glands; almost half its thickness is represented by numerous mucous glands that lie between the muscles and the oral surface of the soft palate. The latter is covered by a stratified squamous epithelium, while the nasal surface is covered with a ciliated columnar epithelium. Taste buds are found on the oral aspect of the soft palate.

In most individuals, two small pits, the fovea palatini, may be seen, one on each side of the midline; they represent the orifices of ducts from some of the minor mucous glands of the palate. In its usual relaxed and pendant position, the anterior (oral) surface of the soft palate is concave and has a median raphe. The posterior aspect is convex and continuous with the nasal floor, the anterosuperior border is attached to the posterior margin of the hard palate, and the sides blend with the pharyngeal wall. The inferior border is free and hangs between the mouth and pharynx. A median conical process, the uvula, projects downwards from its posterior border (see Fig. 31.3). It may be congenitally bifid and associated with submucous cleft palate, hypoplastic orifice of the pharyngotympanic tube and absence of the salpingopharyngeal folds.

The anterior third of the soft palate contains little muscle and consists mainly of the palatine aponeurosis. This region is less mobile and more horizontal than the rest of the soft palate and is the chief area acted on by tensor veli palatini.

A small bony prominence, produced by the pterygoid hamulus, can be felt just behind and medial to each upper alveolar process, in the lateral part of the anterior region of the soft palate. The pterygomandibular raphe (a tendinous band between buccinator and the superior constrictor) passes downwards and outwards from the hamulus to the posterior end of the mylohyoid line (see Fig. 34.8). When the mouth is opened wide, this raphe raises a fold of mucosa that indicates the internal, posterior boundary of the cheek; it is an important landmark for an inferior alveolar nerve block (Ch. 31). (For further reading, see Wood-Jones (1940).)

Palatine aponeurosis A thin, fibrous, palatine aponeurosis, composed of the expanded tendons of the tensor veli palatini muscles, strengthens the soft palate. It is attached to the posterior border and inferior surface of the hard palate behind any palatine crests, and extends medially from behind the greater palatine foramina. It is thick in the anterior two-thirds of the soft palate but very thin further back. Near the midline, it encloses the musculus uvulae. All the other palatine muscles are attached to the aponeurosis.

Palatoglossal and palatopharyngeal arches The lateral wall of the oropharynx presents two prominent folds, the palatoglossal and palatopharyngeal folds (anterior and posterior pillars of the fauces, respectively) (see Figs 31.3, 34.6). The palatoglossal arch, the anterior fold, runs from the soft palate to the side of the tongue and contains palatoglossus. The palatopharyngeal arch, the posterior fold, projects more medially and passes from the soft palate to merge with the lateral wall of the pharynx; it contains palatopharyngeus. A triangular tonsillar fossa (tonsillar sinus) lies on each side of the oropharynx between the diverging palatopharyngeal and palatoglossal arches, and contains the palatine tonsil.

Vascular supply The arterial supply of the soft palate is usually derived from the ascending palatine branch of the facial artery. Sometimes, this is replaced or supplemented by a branch of the ascending pharyngeal artery, which descends forwards between the superior border of the superior constrictor and levator veli palatini, and accompanies the latter to the soft palate. The veins of the soft palate usually drain to the pterygoid venous plexus.

Innervation General sensation from most of the soft palate is carried by branches of the lesser palatine nerve (a branch of the maxillary nerve) and from the posterior part of the palate by pharyngeal branches from the glossopharyngeal nerve and from the plexus around the tonsil (formed by tonsillar branches of the glossopharyngeal and lesser palatine nerves). The special sensation of taste from taste buds in the oral surface of the soft palate is carried in the lesser palatine nerve; the taste fibres initially travel in the greater petrosal nerve (a branch of the facial nerve) and pass through the pterygopalatine ganglion without synapsing. The lesser palatine nerve also carries the secretomotor supply to most of the mucosa of the soft palate, via postganglionic branches from the pterygopalatine ganglion. Postganglionic secretomotor parasympathetic fibres may pass to the posterior parts of the soft palate from the otic ganglion (which receives preganglionic fibres via the lesser petrosal branch of the glossopharyngeal nerve). Postganglionic sympathetic fibres run from the carotid plexus along arterial branches supplying the palate.

Uvulopalatopharyngoplasty

Available with the *Gray's Anatomy* e-book

The pharyngeal airway is kept patent in awake individuals by the combined dilating action of genioglossus, tensor veli palatini, geniohyoid and stylohyoid, which act to counter the negative pressure generated in the lumen of the pharynx during inspiration. The tone in the muscles is reduced during sleep, and is also affected by alcohol and other sedatives, hypothyroidism and a variety of neurological disorders. If the dilator muscle tone is insufficient, the walls of the pharynx may become apposed. Intermittent pharyngeal obstruction may cause snoring, and complete obstruction may cause apnoea, hypoxia and hypercarbia, which lead to arousal and sleep disturbance. Obstructive sleep apnoea (OSA) is a serious syndrome affecting up to 4% of middle-aged people; it has potentially life-threatening consequences. The pathophysiological causes of OSA include an anatomically small upper airway, in which augmented pharyngeal dilator muscle activation can maintain airway patency when the individual is awake but not when asleep, coupled with instability in the respiratory control system. Individuals with OSA may demonstrate some or all of the following anatomical features: an increase in soft palate length and thickness; a shortened mandible; a lowered hyoid bone position; and increases in the volume of fat in soft palate and parapharyngeal fat pads in the retropalatal and retroglossal region. While these studies differ in their assessment of the relative incidence and significance of these factors, they all confirm that the anatomical changes in OSA have the combined effect of both narrowing and shortening the airway. At the same time, physiological mechanisms that exist to ensure that the airways remain dilated and patent seem to suffer a sleep-induced fall in activity in OSA sufferers. The muscles primarily responsible for maintaining a dilated airway under normal conditions are the muscles affecting the position of the hyoid bone, the infrahyoid and suprahyoid muscles, the muscles of the tongue, especially genioglossus, and the muscles that move the soft palate. (For further reading, see Ison et al (1997), Fogel et al (2004), Johal et al (2007), Malhotra and White (2002).)

Surgical techniques involving reduction in the length of the soft palate, removal of the tonsils and plication of the tonsillar pillars can be used to raise the intrinsic dilating tone in the pharyngeal wall and to reduce the bulk of (and to stiffen) the soft palate. This will reduce the tendency of the soft palate to vibrate and generate noise during periods of incipient collapse of the pharynx. An alternative treatment is to deliver air to the pharynx at above atmospheric pressure via a closely fitting facemask, thus inflating the pharynx and countering its tendency to collapse.

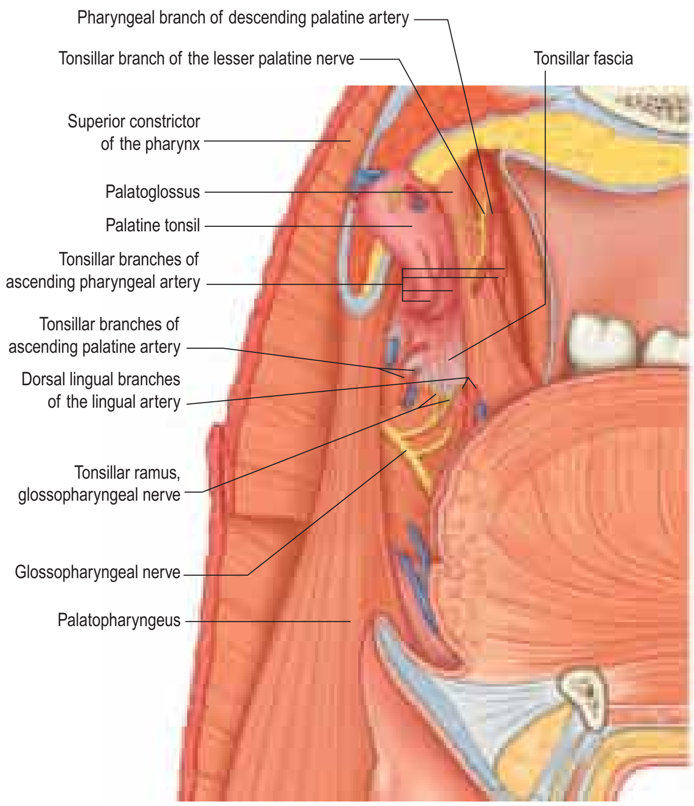

Fig. 34.6 The tonsillar bed, sagittal section.

PALATINE TONSIL

The right and left palatine tonsils form part of the circumpharyngeal lymphoid ring. Each tonsil is an ovoid mass of lymphoid tissue situated in the lateral wall of the oropharynx (**Fig. 34.6**; see Figs 31.3, 31.5). Size varies according to age, individuality and pathological status (tonsils may be hypertrophied and/or inflamed). It is therefore difficult to define the normal appearance of the palatine tonsil. For the first 5 or 6 years of life, the tonsils increase rapidly in size. They usually reach a maximum at puberty, when they average 20–25 mm in vertical, and 10–15 mm in transverse, diameters, and they project conspicuously into the oropharynx. Between the ages of 1 and 11 years, the tonsil and adenoid grow proportionally to the lower facial skeleton (Arens et al 2002). Tonsillar involution begins at puberty, when the reactive lymphoid tissue begins to atrophy, and by old age only a little tonsillar lymphoid tissue remains.

The long axis of the tonsil is directed from above, downwards and backwards. Its medial, free, surface usually presents a pitted appearance. The pits, 10–20 in number, lead into a system of blind-ending, often highly branching, crypts that extend through the whole thickness of the tonsil and almost reach the connective tissue hemicapsule. In a healthy tonsil, the openings of the crypts are fissure-like and the walls of the crypt lumina are collapsed so that they are in contact with each other. The human tonsil is polycryptic. The branching crypt system reaches its maximum size and complexity during childhood. The mouth of a deep tonsillar cleft (intratonsillar cleft, recessus palatinus) opens in the upper part of the medial surface of the tonsil. It is often erroneously called the supratonsillar fossa, and yet it is not situated above the tonsil but within its substance. The mouth of the cleft is semilunar, curving parallel to the convex dorsum of the tongue in the sagittal plane. The upper wall of the recess contains lymphoid tissue that extends into the soft palate as the pars palatina of the palatine tonsil. After the age of 5 years, this embedded part of the tonsil diminishes in size. There is a tendency for the whole tonsil to involute from the age of 14 years, and for the tonsillar bed to flatten out. During young adult life, a mucosal fold, the plica triangularis, stretches back from the palatoglossal arch down to the tongue. It is infiltrated by lymphoid tissue and frequently represents the most prominent (anteroinferior) portion of the tonsil. It rarely persists into middle age.

The lateral or deep surface of the tonsil spreads downwards, upwards and forwards. Inferiorly, it invades the dorsum of the tongue; superiorly, it invades the soft palate; and, anteriorly, it may extend for some distance under the palatoglossal arch. This deep, lateral aspect is covered by a layer of fibrous tissue, the tonsillar hemicapsule. The latter is separable with ease for most of its extent from the underlying muscular wall of the pharynx, which is formed here by the superior constrictor, and sometimes by the anterior fibres of palatopharyngeus, with styloglossus on its lateral side. Anteroinferiorly, the hemicapsule adheres to the side of the tongue and to palatoglossus and palatopharyngeus. In this region, the tonsillar artery, a branch of the facial artery, pierces the superior constrictor to enter the tonsil, accompanied by venae comitantes. An important and sometimes large vein, the external palatine or paratonsillar vein, descends from the soft palate lateral to the tonsillar hemicapsule before piercing the pharyngeal wall. Haemorrhage from this vessel from the upper angle of the tonsillar fossa may complicate tonsillectomy. The muscular wall of the tonsillar fossa separates the tonsil from the ascending palatine artery, and, occasionally, from the tortuous facial artery itself, which may lie near the pharyngeal wall at the lower tonsillar level. The glossopharyngeal nerve lies immediately lateral to the muscular wall of the tonsillar fossa (see Figs 34.3A, 34.6); it is at risk if the wall is pierced, and is commonly temporarily affected by oedema following tonsillectomy. The internal carotid artery lies approximately 25 mm behind and lateral to the tonsil. In some individuals, the styloid process may be elongated and deviate towards the tonsillar bed.

Microstructure

Each tonsil is a mass of lymphoid tissue associated with the oropharyngeal mucosa and fixed in its position, unlike most other examples of mucosa-associated lymphoid tissue. It is covered on its oropharyngeal aspect by non-keratinized stratified squamous epithelium. The whole of the tonsil is supported internally by a delicate meshwork of fine collagen type III (reticulin) fibres which are condensed in places to form more robust connective tissue septa that also contain elastin. These septa partition the tonsillar parenchyma, and merge at their ends with the dense irregular fibrous hemicapsule on the deep aspect of the tonsil and with the lamina propria on the pharyngeal surface. Blood vessels, lymphatics and nerves branch or join within the connective tissue condensations. The hemicapsule forms its lateral boundary with the oropharyngeal wall, and with the mucosa that covers its highly invaginated free surface.

The 10–20 crypts formed by invagination of the free surface mucosa are narrow tubular epithelial diverticula that often branch within the tonsil and frequently are packed with plugs of shed epithelial cells, lymphocytes and bacteria, which may calcify. The epithelium lining the crypts is mostly similar to that of the oropharyngeal surface, i.e. stratified squamous, but there are also patches of reticulated epithelium, which is much thinner, and which has a complex structure that is of great importance in the immunological function of the tonsil.

Reticulated epithelium Reticulated epithelium lacks the orderly laminar structure of stratified squamous epithelium. Its base is deeply invaginated in a complex manner so that the epithelial cells, with their slender branched cytoplasmic processes, provide a coarse mesh to accommodate the infiltrating lymphocytes and macrophages. The basal lamina of this epithelium is discontinuous. Although the oropharyngeal surface is unbroken, the epithelium may become exceedingly thin in places, so that only a tenuous cytoplasmic layer separates the pharyngeal lumen from the underlying lymphocytes. Epithelial cells are held together by small desmosomes, anchored into bundles of keratin filaments. Interdigitating dendritic cells (antigen-presenting cells, APCs) are also present (Ch. 4). The intimate association of epithelial cells and lymphocytes facilitates the direct transport of antigen from the external environment to the tonsillar lymphoid cells, i.e. reticulated epithelial cells are functionally similar to the microfold (M) cells of the gut. The total surface area of the reticulated epithelium is very large because of the complex branched nature of the tonsillar crypts, and has been estimated at 295 cm^2 for an average palatine tonsil.

Tonsillar lymphoid tissue There are four lymphoid compartments in the palatine tonsils. Lymphoid follicles (Ch. 4), many with germinal centres, are arranged in rows roughly parallel to neighbouring connective tissue septa. Their size and cellular content varies in proportion to the immunological activity of the tonsil. The mantle zones of the follicles, each with closely packed small lymphocytes, form a dense cap, always situated on the side of the follicle nearest to the mucosal surface. These cells are the products of B-lymphocyte proliferation within the germinal centres. Extrafollicular, or T-lymphocyte, areas contain a specialized microvasculature including high endothelial venules (HEVs), through which circulating lymphocytes enter the tonsillar parenchyma. The lymphoid tissue of the reticulated crypt epithelium contains predominantly IgG- and IgA-producing B lymphocytes (including some

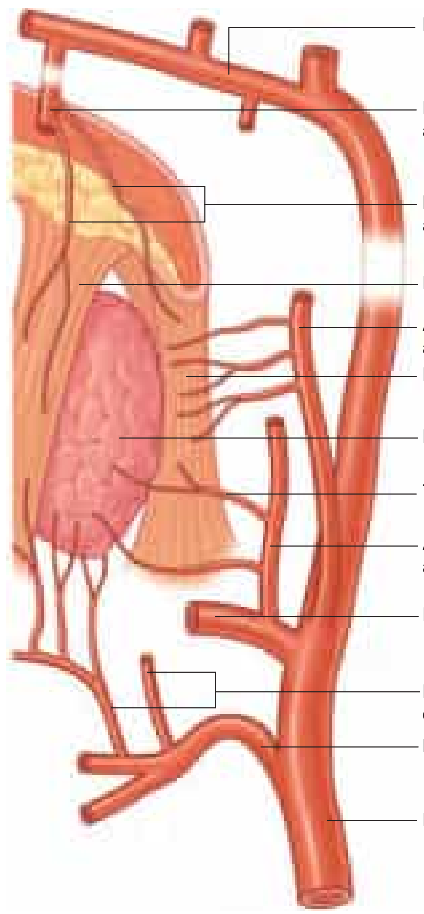

Fig. 34.7 The arterial supply to the palatine tonsil.

mature plasma cells), T lymphocytes and antigen-presenting cells. There are numerous capillary loops in this subsurface region.

Vascular supply and lymphatic drainage

The arterial blood supply to the palatine tonsil is derived from branches of the external carotid artery (see Fig. 34.6; **Fig. 34.7**). Three arteries enter the tonsil at its lower pole. The largest is the tonsillar artery, which is a branch of the facial, or sometimes the ascending palatine, artery. It ascends between medial pterygoid and styloglossus, perforates the superior constrictor at the upper border of styloglossus, and ramifies in the tonsil and posterior lingual musculature. The other arteries found at the lower pole are the dorsal lingual branches of the lingual artery, which enter anteriorly, and a branch from the ascending palatine artery, which enters posteriorly to supply the lower part of the palatine tonsil. The upper pole of the tonsil also receives branches from the ascending pharyngeal artery, which enter the tonsil posteriorly, and from the descending palatine artery and its branches, the greater and lesser palatine arteries. All of these arteries enter the deep surface of the tonsil, branch within the connective tissue septa, narrow to become arterioles and then give off capillary loops into the follicles, interfollicular areas and the cavities within the base of the reticulated epithelium. The capillaries rejoin to form venules, many with high endothelia, and the veins return within the septal tissues to the hemicapsule as tributaries of the pharyngeal drainage. The tonsillar artery and its venae comitantes often lie within the palatoglossal fold, and may haemorrhage if this fold is damaged during surgery.

Unlike lymph nodes, the palatine tonsils do not possess afferent lymphatics or lymph sinuses. Instead, dense plexuses of fine lymphatic vessels surround each follicle and form efferent lymphatics, which pass towards the hemicapsule, pierce the superior constrictor, and drain to the upper deep cervical lymph nodes directly (especially the jugulodigastric nodes) or indirectly through the retropharyngeal lymph nodes. The jugulodigastric nodes are typically enlarged in tonsillitis, when they project beyond the anterior border of sternocleidomastoid and are palpable superficially 1–2 cm below the angle of the mandible; when enlarged, they represent the most common swelling in the neck.

Innervation

The tonsillar region is innervated by tonsillar branches of the maxillary and glossopharyngeal nerves (see Fig. 34.6). The fibres from the maxillary nerve pass through, but do not synapse in, the pterygopalatine ganglion; they are distributed through the lesser palatine nerves and form a plexus (the circulus tonsillaris) around the tonsil together with the tonsillar branches of the glossopharyngeal nerve. Nerve fibres from this plexus are also distributed to the soft palate and the region of the oropharyngeal isthmus. The tympanic branch of the glossopharyngeal nerve supplies the mucous membrane lining the tympanic cavity. Infection, malignancy and postoperative inflammation of the tonsil and tonsillar fossa may therefore be accompanied by pain referred to the ear.

Tonsillectomy

Surgical removal of the pharyngeal tonsils is commonly performed to prevent recurrent acute tonsillitis or to treat airway obstruction by hypertrophied or inflamed palatine tonsils. Occasionally, the tonsil may be removed to treat an acute peritonsillar abscess, which is a collection of pus between the superior constrictor and the tonsillar hemicapsule. Many methods have been employed, the most common being dissection in the plane of the fibrous hemicapsule, followed by ligation or electrocautery to the vessels divided during the dissection. The nerve supply to the tonsil is so diffuse that tonsillectomy under local anaesthesia is performed successfully by local infiltration rather than by blocking the main nerves. Surgical access to the glossopharyngeal nerve may be achieved by separating the fibres of superior constrictor.

Waldeyer's ring

Waldeyer's ring is a circumpharyngeal ring of mucosa-associated lymphoid tissue that surrounds the openings into the digestive and respiratory tracts. It is made up anteroinferiorly by the lingual tonsil, laterally by the palatine and tubal tonsils, and posterosuperiorly by the pharyngeal tonsil and smaller collections of lymphoid tissue in the intertonsillar intervals.

LARYNGOPHARYNX

Boundaries

The laryngopharynx is situated behind the entire length of the larynx (known clinically as the hypopharynx) and extends from the superior border of the epiglottis, where it is delineated from the oropharynx by the lateral glossoepiglottic folds, to the inferior border of the cricoid cartilage, where it becomes continuous with the oesophagus (see Figs 34.1–34.2). The laryngeal inlet lies in the upper part of its incomplete anterior wall, and the posterior surfaces of the arytenoid and cricoid cartilages lie below this opening.

Piriform fossa A small piriform fossa lies on each side of the laryngeal inlet, bounded medially by the aryepiglottic fold and laterally by the thyroid cartilage and thyrohyoid membrane. Branches of the internal laryngeal nerve lie beneath its mucous membrane. At rest, the laryngopharynx extends posteriorly from the lower part of the third cervical vertebral body to the upper part of the sixth. During deglutition, it may be elevated considerably by the hyoid elevators.

Inlet of larynx The obliquely sloping inlet of the larynx lies in the anterior part of the laryngopharynx and is bounded above by the epiglottis, below by the arytenoid cartilages of the larynx, and laterally by the aryepiglottic folds (see Fig. 34.1). Below the inlet, the anterior wall of the laryngopharynx is formed by the posterior surface of the cricoid cartilage.

PHARYNGEAL FASCIA

The two named layers of fascia in the pharynx are the pharyngobasilar and buccopharyngeal fascia. The fibrous layer that supports the pharyngeal mucosa is thickened above the superior constrictor to form the pharyngobasilar fascia (see Fig. 34.10). It is attached to the basilar part of the occipital bone and the petrous part of the temporal bone medial to the pharyngotympanic tube, and to the posterior border of the medial pterygoid plate and the pterygomandibular raphe. Inferiorly, it diminishes in thickness but is strengthened posteriorly by a fibrous band attached to the pharyngeal tubercle of the occipital bone, which descends as the median pharyngeal raphe of the constrictors. This fibrous layer is really the internal epimysial covering of the muscles and their aponeurotic attachment to the base of the skull. The thinner, external part of the epimysium is the buccopharyngeal fascia, which covers the superior constrictor and passes forwards over the pterygomandibular raphe to cover buccinator.

PHARYNGEAL TISSUE SPACES

Pharyngeal tissue spaces can be subdivided into peripharyngeal and intrapharyngeal spaces. The anterior part of the peripharyngeal space is formed by the submandibular and submental spaces, posteriorly by the retropharyngeal space and laterally by the parapharyngeal spaces. The retropharyngeal space is an area of loose connective tissue that lies behind the pharynx and anterior to the prevertebral fascia, extending upwards to the base of the skull and downwards to the retrovisceral space in the infrahyoid part of the neck. Each parapharyngeal space passes laterally around the pharynx and is continuous with the retropharyngeal space. However, unlike the retropharyngeal space, it is a space that is restricted to the suprahyoid region. It is bounded medially by the pharynx, laterally by the pterygoid muscles (where it is part of the infratemporal fossa) and by the sheath of the parotid gland, superiorly by the base of the skull, and inferiorly by suprahyoid structures, particularly the sheath of the submandibular gland; it may be helpful to think of it as shaped like an inverted pyramid extending from the base of the skull to the greater cornu of the hyoid bone. The parapharyngeal space is divided into an anterior, or prestyloid, compartment and a posterior, or retrostyloid, compartment (Maran et al 1984). The prestyloid compartment contains the retromandibular portion of the parotid gland, fat and lymph nodes. The retrostyloid compartment contains the internal carotid artery, the internal jugular vein, the glossopharyngeal, vagus, accessory and hypoglossal nerves, the sympathetic chain, fat and lymph nodes. Any of these structures may be damaged by penetrating injuries directed posterolaterally in the region; more lateral injuries may result in penetration of the parotid gland.

An intrapharyngeal space potentially exists between the inner surface of the constrictor muscles and the pharyngeal mucosa. Infections in this space either are restricted locally or spread through the pharynx into the retropharyngeal or parapharyngeal spaces. The peritonsillar space is an important part of the intrapharyngeal space; it lies around the palatine tonsil between the pillars of the fauces. Infections in the intratonsillar space usually spread up or down the intrapharyngeal space, or through the pharynx into the parapharyngeal space.

Tissue spaces between the layers of cervical fascia are described on page 446; tissue spaces around the larynx are described on page 594.

SPREAD OF INFECTION

Infection that spreads into the parapharyngeal space will produce pain and trismus. There may be swelling in the oropharynx that extends up to the uvula, displacing it to the contralateral side, and dysphagia. Posterior spread from the parapharyngeal space into the retropharyngeal space will produce bulging of the posterior pharyngeal wall, dyspnoea and nuchal rigidity. Involvement of the carotid sheath may produce symptoms caused by thrombosis of the internal jugular vein and cranial nerve symptoms involving the glossopharyngeal, vagus, accessory and hypoglossal nerves. If the infection continues to spread unchecked, mediastinitis will ensue. A virulent infection in the retropharyngeal space may spread through the prevertebral fascia into the underlying danger space; infection in this tissue space may descend into the thorax and even below the diaphragm, and results in chest pain, severe dyspnoea and retrosternal discomfort.

Pharyngeal infection from mucosa-associated lymph tissues such as the palatine tonsil, or as a result of a penetrating injury (e.g. from an ingested foreign body), may result in the spread of infection into the tissue spaces of the neck adjacent to the pharynx. This is an extremely serious situation because there is potential for rapid spread throughout the neck and, more dangerously, to the superior mediastinum, to cause overwhelming life-threatening infection.

PARAPHARYNGEAL SPACE TUMOURS

Tumours that develop in the parapharyngeal tissue space may remain asymptomatic for some time. When they do present, it may be with a diffuse pattern of symptoms, reflecting the effects of compression on the lower cranial nerves, e.g. dysarthria, resulting from impairment of tongue movements secondary to hypoglossal nerve damage; dysphagia, with overspill and aspiration of ingested material into the airway, resulting from loss of sensory information from the territory of the pharyngeal plexus nerves; motor dysfunction of the pharynx and larynx, resulting from loss of motor innervation via the pharyngeal plexus and the recurrent laryngeal branch of the vagus to the intrinsic muscles of the larynx; and voice changes reflecting involvement of the laryngeal branches of the vagus. These tumours may also give rise to snoring as a result of narrowing of the nasopharynx.

Several surgical approaches have been described for the management of parapharyngeal space tumours, including transcervical, transparotid, transcervical–transmandibular and transoral approaches. Transoral robotic surgery uses the oral cavity as a surgical corridor; as yet, there have been relatively few studies from the transoral perspective of the relevant surgical anatomy (Dallan et al 2011, Moore et al 2012, Wang et al 2014).

MUSCLES OF THE SOFT PALATE AND PHARYNX

The muscles of the soft palate and pharynx are levator veli palatini, tensor veli palatini, palatoglossus, palatopharyngeus, musculus uvulae, salpingopharyngeus, stylopharyngeus, and the superior, middle and inferior constrictors.

Levator veli palatini

Levator veli palatini arises by a small tendon from a quadrilateral roughened area on the medial end of the inferior surface of the petrous part of the temporal bone, in front of the lower opening of the carotid canal (see Fig. 34.3; **Figs 34.8–34.10**). Additional fibres arise from the inferior aspect of the cartilaginous part of the pharyngotympanic tube and from the vaginal process of the sphenoid bone. At its origin, the muscle is inferior rather than medial to the pharyngotympanic tube and only crosses medial to it at the level of the medial pterygoid plate. It passes medial to the upper margin of the superior constrictor and

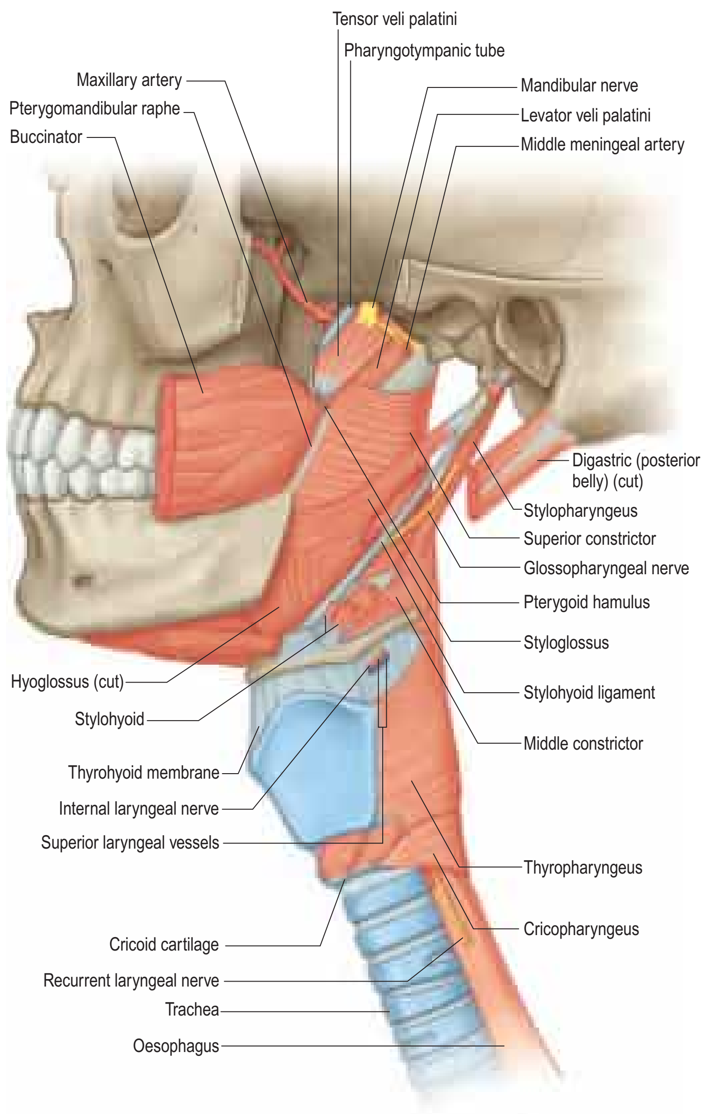

Fig. 34.8 Muscles of the pharynx, lateral view.

SECTION 4

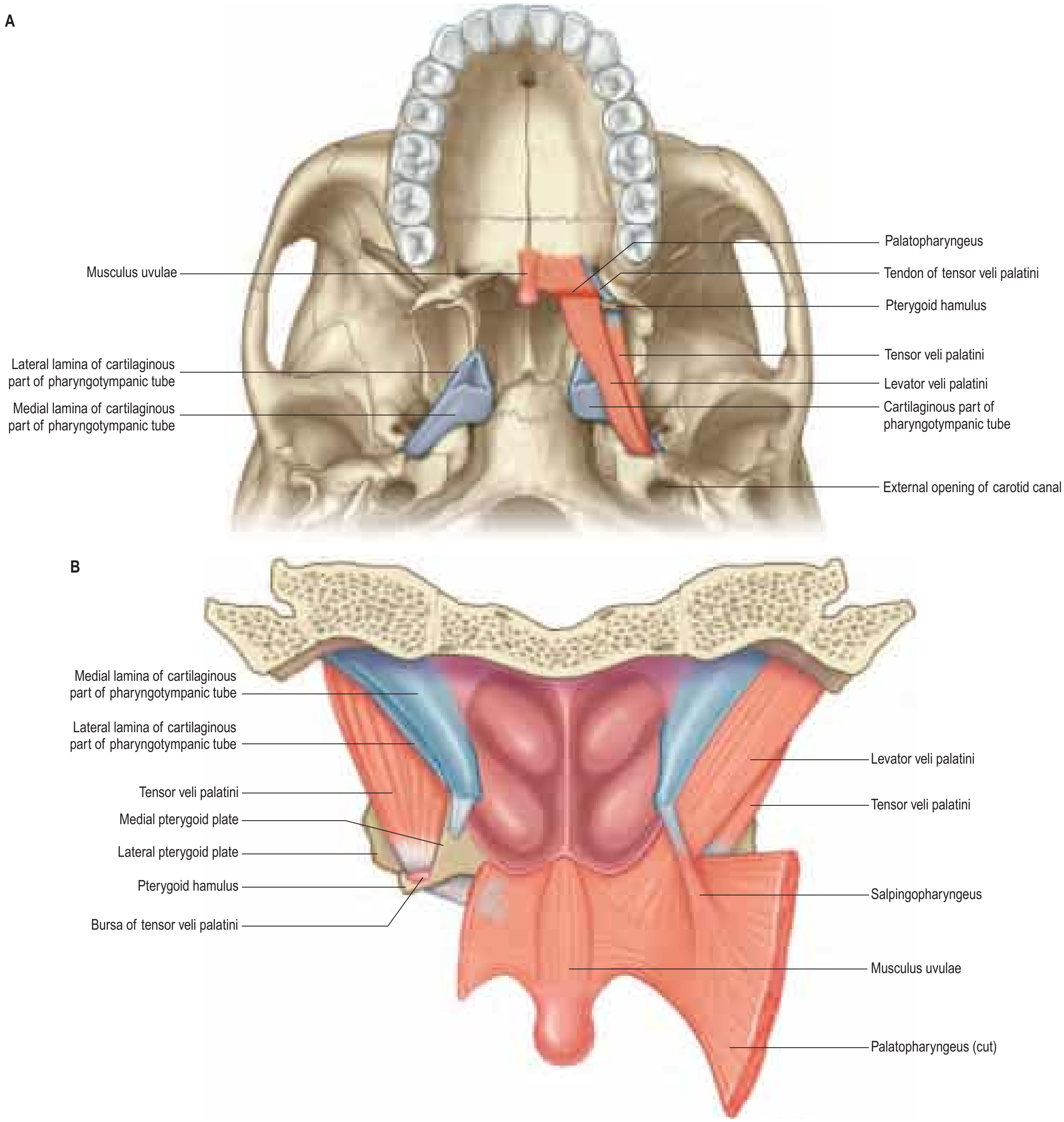

Fig. 34.9 Muscles of the soft palate. **A**, Inferior view. **B**, Superior view.

anterior to salpingopharyngeus. Its fibres spread in the medial third of the soft palate between the two strands of palatopharyngeus to attach to the upper surface of the palatine aponeurosis as far as the midline, where they interlace with those of the contralateral muscle. Thus, the two levator muscles form a sling above and just behind the palatine aponeurosis.

Vascular supply The blood supply of levator veli palatini is derived from the ascending palatine branch of the facial artery and the greater palatine branch of the maxillary artery (Freelander 1992).

Innervation Levator veli palatini is innervated via the pharyngeal plexus.

Actions The primary role of the levator veli palatini muscles is to elevate the almost vertical posterior part of the soft palate and pull it slightly backwards; during swallowing, the soft palate is elevated so that it touches the posterior wall of the pharynx, separating the nasopharynx from the oropharynx. By additionally pulling on the lateral walls of the nasopharynx posteriorly and medially, the levator veli palatini muscles also narrow that space. The muscle has little or no effect on the pharyngotympanic tube, although it might allow passive opening.

Tensor veli palatini

Tensor veli palatini arises from the scaphoid fossa of the pterygoid process and posteriorly from the medial aspect of the spine of the sphenoid bone (see Figs 32.3A, B, 34.3, 34.8–34.10). Between these two sites, it is attached to the anterolateral membranous wall of the pharyngotympanic tube (including its narrow isthmus where the cartilaginous medial two-thirds meets the bony lateral third). Some fibres may be continuous with those of tensor tympani. Inferiorly, the fibres converge on a delicate tendon that turns medially around the pterygoid hamulus to pass through the attachment of buccinator to the palatine aponeurosis and the osseous surface behind the palatine crest on the horizontal plate of the palatine bone. There is a small bursa between the tendon and the pterygoid hamulus (see Fig. 131-8 in Cummings Otolaryngology Head & Neck Surgery 5e).

The muscle is thin and triangular, and lies lateral to the medial pterygoid plate, pharyngotympanic tube and levator veli palatini. Its lateral surface contacts the upper and anterior part of medial pterygoid, the mandibular, auriculotemporal and chorda tympani nerves, the otic ganglion and the middle meningeal artery.

Dilator tubae Some fibres of tensor veli palatini that arise from the hamulus of the medial pterygoid plate are attached to the short lateral

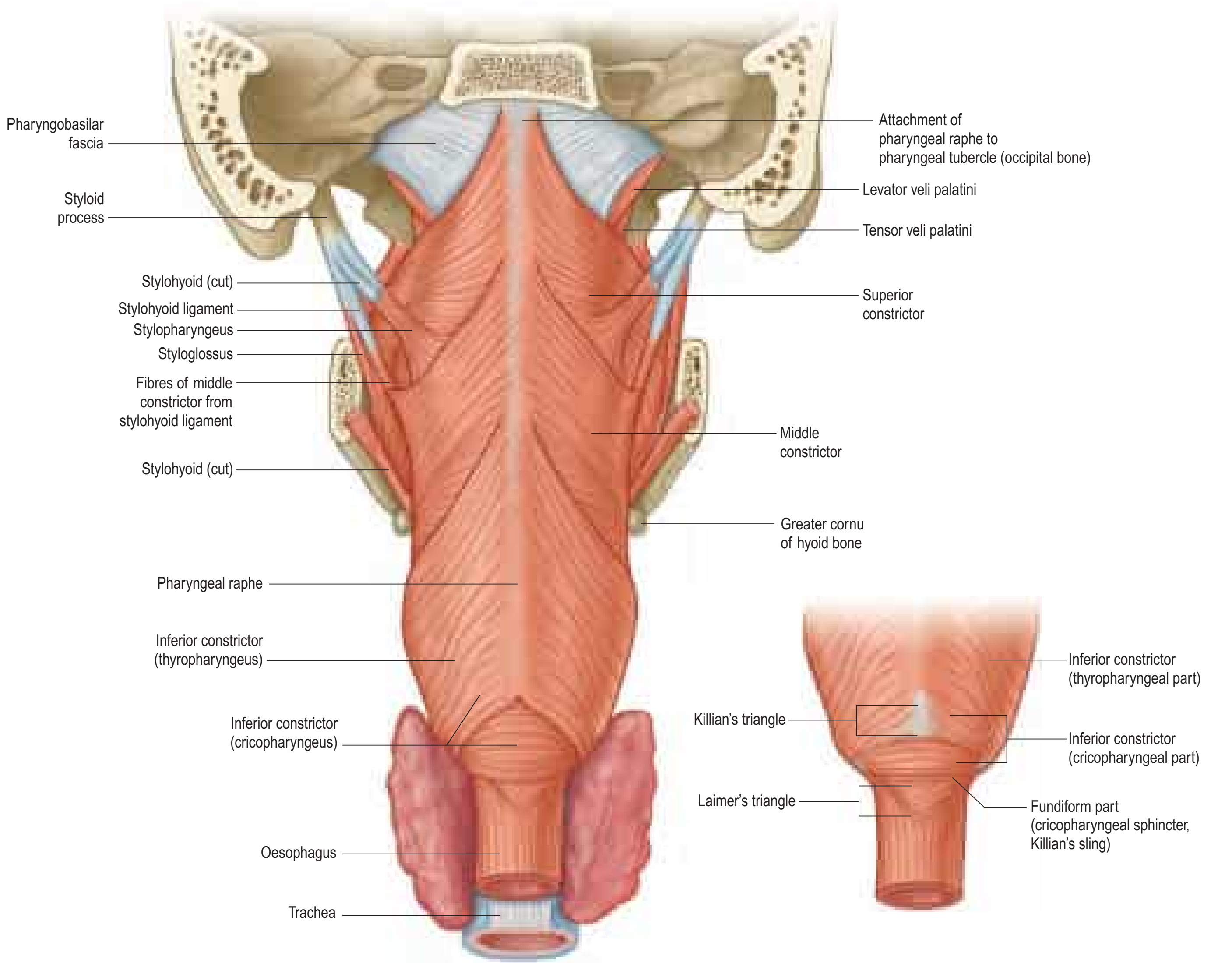

Fig. 34.10 Muscles of the pharynx and the pharyngobasilar fascia, posterior view.

lamina of the cartilage of the pharyngotympanic, to a condensation of connective tissue lateral to the tubal wall and to a portion of Ostmann's fat pad; these fibres are sometimes referred to as dilator tubae.

Vascular supply The blood supply of tensor veli palatini is derived from the ascending palatine branch of the facial artery and the greater palatine branch of the maxillary artery (Freelander 1992).

Innervation The motor innervation of tensor veli palatini is derived from the mandibular nerve via the nerve to medial pterygoid, and reflects the development of the muscle from the first branchial arch.

Actions Tensor veli palatini is said to have three anchoring points on which it isometrically contracts: the pterygoid hamulus, Ostmann's fat pad and medial pterygoid (O'Reilly and Sando 2010). Acting together, the two tensor veli palatini muscles tauten the soft palate, principally its anterior part, and depress it by flattening its arch. Acting unilaterally, the muscle pulls the soft palate to one side. Although contraction of both muscles will slightly depress the anterior part of the soft palate, it is often assumed that the increased rigidity aids palatopharyngeal closure. A primary role of the tensor is to open the pharyngotympanic tube, e.g. during deglutition and yawning, via its dilator tubae component, and so aid equalization of air pressure between the middle ear and nasopharynx.

Palatoglossus

Palatoglossus is narrower at its middle than at its ends (see Figs 34.3, 34.6). Together with its overlying mucosa, it forms the palatoglossal arch or fold (see Fig. 31.3). It arises from the oral surface of the palatine aponeurosis, where it is continuous with its contralateral fellow. It extends forwards, downwards and laterally in front of the palatine tonsil to the side of the tongue. Some of its fibres spread over the dorsum of the tongue, while others pass deeply into its substance to intermingle with fibres of the intrinsic transverse muscle.

Vascular supply Palatoglossus receives its blood supply from the ascending palatine branch of the facial artery and from the ascending pharyngeal artery.

Innervation Palatoglossus is innervated via the pharyngeal plexus, and is therefore unlike all the other muscles of the tongue, which are supplied by the hypoglossal nerve.

Actions Palatoglossus elevates the root of the tongue and approximates the palatoglossal arch to its contralateral fellow, thus shutting off the oral cavity from the oropharynx.

Palatopharyngeus

Palatopharyngeus and its overlying mucosa form the palatopharyngeal arch (see Fig. 31.3). Within the soft palate, palatopharyngeus is composed of two fasciculi that are attached to the upper surface of the palatine aponeurosis; they lie in the same plane but are separated from each other by levator veli palatini (see Figs 34.3, 34.9). The thicker, anterior fasciculus arises from the posterior border of the hard palate as well as the palatine aponeurosis, where some fibres interdigitate across the midline. The posterior fasciculus is in contact with the mucosa of the pharyngeal aspect of the palate, and joins the posterior band of the contralateral muscle in the midline. The two layers unite at the posterolateral border of the soft palate, and are joined by fibres of salpingopharyngeus. Passing laterally and downwards behind the tonsil, palatopharyngeus descends posteromedial to and in close contact with stylopharyngeus, to be attached with it to the posterior border of the thyroid cartilage. Some fibres end on the side of the pharynx, attached to pharyngeal fibrous tissue, and others cross the midline posteriorly, decussating with those of the contralateral muscle. Palatopharyngeus thus forms an incomplete internal longitudinal muscular layer in the wall of the pharynx.

Passavant's muscle (palatopharyngeal sphincter) The existence of Passavant's muscle remains controversial (Cho et al 2013). It has been described as a part of the superior constrictor and palatopharyngeus muscles (see Fig. 34.3B). An alternative view holds that it is a distinct palatine muscle that arises from the anterior and lateral parts of the upper surface of the palatine aponeurosis, lies lateral to levator veli palatini, blends internally with the upper border of the superior constrictor, and encircles the pharynx as a sphincter-like muscle. Whatever its origin, when it contracts, it forms a ridge (Passavant's ridge) when the soft palate is elevated. The change from columnar, ciliated, 'respiratory' epithelium to stratified, squamous epithelium that takes place on the superior aspect of the soft palate occurs along the line of attachment of the palatopharyngeal sphincter to the palate. The muscle is hypertrophied in cases of complete cleft palate.

Vascular supply Palatopharyngeus receives its arterial supply from the ascending palatine branch of the facial artery, the greater palatine branch of the maxillary artery and the pharyngeal branch of the ascending pharyngeal artery.

Innervation Palatopharyngeus is innervated via the pharyngeal plexus.

Actions Acting together, the palatopharyngei pull the pharynx up, forwards and medially, and thus shorten it during swallowing. They also approximate the palatopharyngeal arches and draw them forwards.

Musculus uvulae

Musculus uvulae arises from the posterior nasal spine of the palatine bone and the superior surface of the palatine aponeurosis, and lies between the two laminae of the aponeurosis (see Figs 34.1, 34.9). It runs posteriorly above the sling formed by levator veli palatini and inserts beneath the mucosa of the uvula. The two sides of the muscle are united along most of its length.

Vascular supply The blood supply of musculus uvulae is derived from the ascending palatine branch of the facial artery and the descending palatine branch of the maxillary artery.

Innervation The nerve supply to musculus uvulae is innervated via the pharyngeal plexus.

Actions By retracting the uvular mass and thickening the middle third of the soft palate, musculus uvulae aids levator veli palatini in palatopharyngeal closure. The two muscles run at right angles to each other and their contraction raises a 'levator eminence' that helps seals off the nasopharynx.

Salpingopharyngeus

Salpingopharyngeus arises from the inferior part of the cartilage of the pharyngotympanic tube near its pharyngeal opening and passes downwards within the salpingopharyngeal fold to blend with palatopharyngeus (see Figs 34.1, 34.3A, 34.9).

Vascular supply Salpingopharyngeus receives its arterial supply from the ascending palatine branch of the facial artery, the greater palatine branch of the maxillary artery and the pharyngeal branch of the ascending pharyngeal artery.

Innervation Salpingopharyngeus is innervated via the pharyngeal plexus.

Actions Salpingopharyngeus elevates the pharynx, and may also assist tensor veli palatini to open the cartilaginous end of the pharyngotympanic tube during swallowing.

Stylopharyngeus

Stylopharyngeus is a long slender muscle, cylindrical above and flat below. It arises from the medial side of the base of the styloid process, descends along the side of the pharynx, and passes between the superior and middle constrictors to spread out beneath the mucous membrane (see Figs 34.3, 34.8, 34.10, 31.6). Some fibres merge into the constrictors and the lateral glossoepiglottic fold, while others join fibres of palatopharyngeus and are attached to the posterior border of the thyroid cartilage. The glossopharyngeal nerve curves round the posterior border and the lateral side of stylopharyngeus, and passes between the superior and middle constrictors to reach the tongue.

Vascular supply Stylopharyngeus receives its arterial supply from the pharyngeal branch of the ascending pharyngeal artery.

Innervation Stylopharyngeus is innervated by the glossopharyngeal nerve.

Actions Stylopharyngeus elevates the pharynx and larynx.

Superior constrictor

The superior constrictor is a quadrilateral sheet of muscle and is thinner than the other two constrictors. It is attached anteriorly to the pterygoid hamulus (and sometimes to the adjoining posterior margin of the medial pterygoid plate), the posterior border of the pterygomandibular raphe, the posterior end of the mylohyoid line of the mandible and, by a few fibres, to the side of the tongue (see Figs 34.1, 34.3, 34.6, 34.8, 34.10, 31.6). The fibres curve back into a median pharyngeal raphe that is attached superiorly to the pharyngeal tubercle on the basilar part of the occipital bone.

Relations The upper border of the superior constrictor is separated from the cranial base by a crescentic interval that contains levator veli palatini, the pharyngotympanic tube and an upward projection of pharyngobasilar fascia. The lower border is separated from the middle constrictor by stylopharyngeus and the glossopharyngeal nerve (see Fig. 34.8). Anteriorly, the pterygomandibular raphe separates the superior constrictor from buccinator, and, posteriorly, the superior constrictor lies on the prevertebral muscles and fascia, from which it is separated by the retropharyngeal space. The ascending pharyngeal artery, pharyngeal venous plexus, glossopharyngeal and lingual nerves, styloglossus, middle constrictor, medial pterygoid, stylopharyngeus and the stylohyoid ligament all lie laterally, and palatopharyngeus, the tonsillar capsule and the pharyngobasilar fascia lie internally.

Vascular supply The arterial supply of the superior constrictor is derived mainly from the pharyngeal branch of the ascending pharyngeal artery and the tonsillar branch of the facial artery.

Innervation The superior constrictor is innervated via the pharyngeal plexus.

Actions The superior constrictor constricts the upper part of the pharynx.

Middle constrictor

The middle constrictor is a fan-shaped sheet attached anteriorly to the lesser cornu of the hyoid and the lower part of the stylohyoid ligament (the chondropharyngeal part of the muscle), and to the whole of the upper border of the greater cornu of the hyoid (the ceratopharyngeal part) (see Figs 34.3A, 34.8, 34.10, 31.6). The lower fibres descend deep to the inferior constrictor to reach the lower end of the pharynx; the middle fibres pass transversely, and the superior fibres ascend and overlap the superior constrictor. All fibres insert posteriorly into the median pharyngeal raphe (Sakamoto 2014).

Relations The glossopharyngeal nerve and stylopharyngeus pass through a small gap between the middle and superior constrictors, and the internal laryngeal nerve and the laryngeal branch of the superior thyroid artery pass between the middle and inferior constrictors. The prevertebral fascia and longus colli and longus capitis are posterior; the superior constrictor, stylopharyngeus and palatopharyngeus are internal; and the carotid vessels, pharyngeal plexus of nerves and some lymph nodes are lateral. Near its hyoid attachment, the middle constrictor lies deep to hyoglossus, from which it is separated by the lingual artery.

Vascular supply The arterial supply of the middle constrictor is derived mainly from the pharyngeal branch of the ascending pharyngeal artery and the tonsillar branch of the facial artery.

Innervation The middle constrictor is innervated via the pharyngeal plexus.

Actions The middle constrictor constricts the middle part of the pharynx during swallowing.

Inferior constrictor

The inferior constrictor is the thickest of the three constrictor muscles and is usually described in two parts: thyropharyngeus and cricopharyngeus (see Figs 34.3A, 34.8, 34.10, 31.6). Thyropharyngeus arises from the oblique line of the thyroid lamina, a strip of the lamina behind this, and by a small slip from the inferior cornu. Some additional fibres arise from a tendinous cord that loops over cricothyroid. Cricopharyngeus arises from the side of the cricoid cartilage between the attachment of cricothyroid and the articular facet for the inferior thyroid cornu. Some authors have described cricopharyngeus as consisting of a superficial upper oblique portion – the pars oblique – and a lower, deeper, transverse portion – the pars fundiformis. The upper part attaches to the median raphe while the lower part forms a circular band that lacks a median raphe. The area demarcated by the pars oblique and pars fundiformis of cricopharyngeus is termed Killian's dehiscence (or Killian's triangle). A second triangular area, Laimer's triangle, can be identified beneath cricopharyngeus between the longitudinal fibres of the oesophagus as they pass laterally on either side to attach to the cricoid cartilage; only the circular muscle of the oesophagus forms the wall here. Both triangles are postulated to be sites of weakness in the wall of the pharynx and oesophagus, and are therefore areas where diverticula could potentially form. Both cricopharyngeus and thyropharyngeus spread posteromedially to join the contralateral muscle. Thyropharyngeus is inserted into the median pharyngeal raphe, and its upper fibres ascend obliquely to overlap the middle constrictor; however, cricopharyngeus blends with the circular oesophageal fibres around the narrowest part of the pharynx.

Relations The buccopharyngeal fascia is external; the prevertebral fascia and muscles are posterior; the thyroid gland, common carotid artery and sternothyroid are lateral; and the middle constrictor, stylopharyngeus, palatopharyngeus and the fibrous lamina are internal. The internal laryngeal nerve and laryngeal branch of the superior thyroid artery reach the thyrohyoid membrane by passing between the inferior and middle constrictors. The external laryngeal nerve descends on the superficial surface of the muscle, just behind its thyroid attachment, and pierces its lower part. The recurrent laryngeal nerve and the laryngeal branch of the inferior thyroid artery ascend deep to its lower border to enter the larynx.

Vascular supply The arterial supply of the inferior constrictor is derived mainly from the pharyngeal branch of the ascending pharyngeal artery and the muscular branches of the inferior thyroid artery.

Innervation Both parts of the inferior constrictor are usually innervated via the pharyngeal plexus. Although controversial, available evidence in humans suggests that cricopharyngeus is also supplied by the recurrent laryngeal nerve and the external branch of the superior laryngeal nerve (Sakamoto 2013).

Actions Thyropharyngeus constricts the lower part of the pharynx. Cricopharyngeus is the main component of the upper oesophageal sphincter, or pharyngo-oesophageal high-pressure zone, the other parts being thyropharyngeus and the proximal cervical oesophagus. (The extent to which the lower fibres of thyropharyngeus and the upper fibres of the oesophageal musculature are involved in closing the upper end of the oesophagus appears to depend on the physiological state, whereas cricopharyngeus always participates in closure.) The upper oesophageal sphincter is defined manometrically as a region of elevated intraluminal pressure, 2–4 cm long, located at the junction of the hypopharynx and cervical oesophagus.

Cricopharyngeus contains about 40% of endomysial connective tissue, much of which is elastic, but it lacks muscle spindles (Bonnington et al 1988, Brownlow et al 1989). It contains both slow-twitch type I and fast-twitch type II fibres, a structural arrangement that underpins the various functions of the upper oesophageal sphincter, i.e. maintaining constant basal tone, yet being able to relax and contract rapidly during swallowing, belching and vomiting. The tonic activity of cricopharyngeus between swallows prevents influx of air during inspiration and tracheobronchial aspiration and pharyngeal reflux of oesophageal contents during oesophageal peristalsis. For further reading, see Lang and Shaker (2000).

Hypopharyngeal diverticula

Hypopharyngeal diverticula occur in the lower portion of the pharynx through areas of weakness in the pharyngeal wall. The pharyngeal mucosa that lies between cricopharyngeus and thyropharyngeus is relatively unsupported by pharyngeal muscles and is called the dehiscence of Killian. A delay in the relaxation of cricopharyngeus, which can occur when the swallowing mechanism becomes discoordinated, generates a zone of elevated pressure adjacent to the mucosa in the dehiscence. The result is the development of a pulsion diverticulum (a pouch of prolapsing mucosa), which breaches the thin muscle wall adjacent to the sixth cervical vertebra and expands, usually a little to the left side, into the parapharyngeal potential space. This may trap portions (or all) of the passing food bolus, resulting in regurgitation of old food, aspiration pneumonia, halitosis and weight loss. Treatment may involve open excision or inversion of the pouch to prevent it filling, coupled with division of the circular fibres of cricopharyngeus, to prevent the build-up of pressure in the region and recurrence of the pouch.

The majority of hypopharyngeal diverticula arise either between the two parts of cricopharyngeus (where the upper oblique and lower transverse fibres diverge) or below cricopharyngeus (where the longitudinal fibres of the oesophagus diverge in Laimer's triangle). Some arise between cricopharyngeus and thyropharyngeus. Diverticula are usually midline but may arise laterally below cricopharyngeus.

PHARYNGEAL PLEXUS

Almost all of the nerve supply to the pharynx, whether motor or sensory, is derived from the pharyngeal plexus, which is formed by the pharyngeal branches of the glossopharyngeal and vagus nerves with contributions from the superior cervical sympathetic ganglion. The plexus lies on the external surface of the pharynx, especially on the middle constrictor. Filaments from the plexus ascend or descend external to the superior and inferior constrictors before branching within the muscular layer and mucosa of the pharynx.

The pharyngeal branch of the vagus supplies all the muscles of the soft palate (excluding tensor veli palatini, which is supplied by the mandibular division of the trigeminal via the nerve to medial pterygoid) and all the muscles of the pharynx (excluding stylopharyngeus, which is supplied by the glossopharyngeal nerve). It emerges from the upper part of the inferior vagal ganglion and consists of axons arising from neuronal cell bodies in the nucleus ambiguus. The nerve passes between the external and internal carotid arteries to reach the upper border of the middle pharyngeal constrictor and subsequently divides into numerous filaments that contribute to the pharyngeal plexus. It also gives off a minute filament, the ramus lingualis vagi, which joins the hypoglossal nerve as it curves round the occipital artery.

Contemporary evidence does not support the description of a cranial root of the accessory nerve as the conduit for the major motor drive to the pharyngeal plexus, albeit via the vagus nerve. The plexiform connections that may be demonstrated between the accessory nerve and the vagus nerve within the posterior cranial fossa are too variable and insubstantial to support this function.

ANATOMY OF SWALLOWING (DEGLUTITION)

Swallowing involves a series of activities that occur within a matter of seconds. Traditionally described as a reflex, the process is more properly regarded as a programmed motor behaviour. Swallowing is initiated when food or liquid stimulates sensory nerves in the oropharynx. In a 24-hour period, an average person will swallow between 600 and 1000 times but, of these, only some 150 will relate to feeding; the remainder occur to clear continuously produced saliva and are less frequent at night (Sato and Nakashima 2006, 2007).

Eating and drinking are basic human pleasures, and problems associated with swallowing can impact dramatically on the quality of life. Swallowing disorders are usually symptoms of other complex diseases; an inability to swallow may adversely affect nutritional status and therefore indirectly exacerbate the underlying disease. Aside from the risk of asphyxiation through choking, swallowing disorders can also be a direct cause of morbidity and mortality as a result of aspiration of food, liquid or possibly refluxed gastric acid contents, causing bacterial infection or tissue damage.

Swallowing in the adult human has usually been studied in relation to swallowing solid or liquid food carried out on command. For descriptive purposes, the process has been traditionally divided into four phases: oral preparatory, oral transit/transfer, pharyngeal and oesophageal (in this traditional view some have only recognized three phases, combining the two oral phases of preparation and transit into a single oral phase). While this sequence of events may still be appropriate for describing the swallowing of liquids, it does not accurately represent the way in which solid food is prepared for swallowing, where suitably processed food is passed in stages to the oropharynx and valleculae until a swallow is initiated. Oral and pharyngeal stages overlap when swallowing solid food, but perhaps less so when swallowing saliva or other liquids. It therefore seems appropriate to speak of an oral preparatory and an oral transit phase when describing the swallowing of liquids, but inappropriate when considering the swallowing of solid food. (For further reading, see Hiiemae and Palmer (1999, 2003), Matsuo and Palmer (2008), Mioche et al (2002), Palmer et al (2007).)

ORAL PREPARATORY PHASE

In the oral preparatory phase, liquids are taken into the mouth and held there either on the floor of the mouth or against the hard palate by the upward movement of the tongue. During swallowing, the muscles that dilate the pharyngotympanic tube are activated during a pause in respiration in the expiratory phase. Throughout this first phase, the soft palate is fully lowered by contraction of palatoglossus and palatopharyngeus, and the posterior part of the tongue is simultaneously elevated; the apposed soft palate and tongue form a tight seal that helps to prevent premature leakage of the bolus into the oropharynx before the airways are fully protected. Slight leakage of fluid does sometimes occur; the tendency for there to be leakage because of imperfect sealing increases with age.

ORAL TRANSIT/TRANSFER PHASE

In the oral transit/transfer phase, the liquid in the oral cavity is transported through the palatoglossal and palatopharyngeal arches into the oropharynx. Genioglossus raises both the tongue tip and the part of the tongue immediately behind the tip so that they come into contact with the alveolar ridge. Orbicularis oris and buccinator remain contracted, keeping the lips and cheeks taut and the liquid central in the oral cavity. The liquid is accommodated in a shallow midline gutter that forms along the dorsum of the tongue, probably as a result of the co-contraction of the styloglossi and the genioglossi, aided by the superior longitudinal and transverse fibres of the intrinsic muscles. The mandible is elevated and the mouth is closed. The floor of the mouth and the anterior and middle portions of the tongue are elevated by co-contraction of the suprahyoid group of muscles (mylohyoid, digastric, geniohyoid and stylohyoid); the effectiveness of the suprahyoid muscles is increased as they contract against a fixed mandible (the mouth does not have to be closed to swallow, but it is much harder to swallow if it is open). Contraction of stylohyoid elevates the more posterior parts of the tongue and empties the longitudinal gutter. At the same time, the tongue flattens, probably as a result of the contraction of hyoglossus and some of the intrinsic lingual muscles, especially the vertical fibres. The elevated, flattened tongue pushes the liquid against the hard palate, and the sides of the tongue seal against the maxillary alveolar processes, helping to move the liquid further posteriorly. Contraction of styloglossus and mylohyoid completes the elevation of the posterior part of the tongue. At the same time, the posterior oral seal relaxes and the posterior tongue moves forwards; the overall effect is of a cam-like action of the tongue, sweeping or squeezing the liquid towards the pillars of the fauces, finally delivering it to the oropharynx and initiating the pharyngeal stage of swallowing, where the pharyngeal aperture is initially increased and then closed.

ORAL PHASES WHEN SWALLOWING SOLIDS

When solid food is swallowed, the process is slightly different from that just described for swallowing liquids. Food is mixed with saliva and reduced to smaller pieces by the processes of chewing. When it has been converted to a suitable consistency to be swallowed, it is transferred to the oropharynx and valleculae, where it can be retained for a few seconds prior to swallowing. During this time, chewing may continue; the bolus is progressively augmented in one or more stages until a swallow is initiated and the pharyngeal phase begins. In this part of the 'oral phase for solid food', the essential action is chewing; the mandible is moved by the action of the jaw elevators and depressors (Ch. 32), and the food is reduced by the grinding action of the teeth and simultaneously mixed with saliva. The lips are maintained as a tight labial seal by the contraction of orbicularis oris; buccinator performs a similar function for the cheeks. In this way, the sulci are closed, the vestibule normally remains empty, and any food that enters the vestibule is returned to the oral cavity proper. Buccinator keeps the cheeks taut, ensuring that they are kept clear of the occlusal surfaces and that the food remains in place under the occlusal surfaces of the molar teeth. Loss of the nerve supply to buccinator as a result of damage to the facial nerve results in painful and repeated lacerations of the cheeks. It was thought that the soft palate was depressed throughout this phase. It is now recognized that the posterior seal is not tight, and that the oral cavity and oropharynx remain in communication, permitting addition of material to the bolus while chewing continues. Spillage occurs because the soft palate is not in continuous contact with the posterior part of the tongue, as was once thought (Hiiemae and Palmer 1999). Bolus formation appears to involve several cycles of transporting food from the anterior to the posterior part of the tongue through the palatoglossal and palatopharyngeal arches until a bolus accumulates on the oropharyngeal surface of the tongue (retrolingual loading), the valleculae and within the oropharynx. Throughout this phase, the lateral and rotatory tongue movements that deliver the food to the teeth for grinding and reduction are crucial for normal bolus formation because they ensure that the food is positioned under the occlusal surfaces of the teeth. If effective tongue movements do not occur, chewing will be compromised. Movements of the tongue are also cyclical in phase with the movements of the jaw and hyoid bone. As the jaw is depressed, the tongue is also depressed and moves anteriorly. As the jaw is elevated, the tongue is retracted so that it no longer lies under the anterior teeth as they as are brought together by jaw elevation. These coordinated actions help to ensure that the tongue is not usually bitten during chewing.

Chewing continues until all the food has been moved posteriorly, a process that can last from less than 1 second to as much as 10 seconds. The oral preparatory, oral transport and pharyngeal phases of swallowing therefore overlap when solid food is being swallowed. As pieces of food are prepared in the mouth, they are moved to the posterior part of the tongue or on into the oropharynx in a similar manner to that used to transport liquids (see above). The end of this phase of swallowing is marked by the tongue propelling the prepared bolus of food to the posterior part of the oral cavity and then on into the oropharynx to initiate the swallow. Contact with either the posterior wall of the oropharynx or the mucosa overlying the palatoglossal and palatopharyngeal arches was once thought to be necessary to initiate a swallow; it is now known that there is a great degree of variability in the position of the bolus at the time at which a swallow is initiated.

PHARYNGEAL PHASE

The delivery of the bolus to the oropharynx triggers the pharyngeal phase of the swallow. This phase, which is involuntary and is the most critical stage of swallowing, involves the pharynx changing from being an air channel (between the posterior nares and laryngeal inlet) to a food channel (from the fauces to the upper end of the oesophagus). The airway is protected from aspiration during swallowing by hyolaryngeal elevation, and by resetting respiratory rhythm so that airflow ceases briefly as the bolus passes through the hypopharynx; the total time that elapses from the bolus triggering the pharyngeal phase to the re-establishment of the airway is barely 1 second. Thus there are two aspects of the process to be examined: the transport of the food down the pharynx and through the upper oesophageal sphincter, and the absolute need to protect the airways throughout this time.

The nasopharynx is sealed off from the oropharynx by activation of the superior pharyngeal constrictor and contraction of a subset of palatopharyngeal fibres to form a variable, ridge-like structure (Passavant's ridge; Cho et al 2013), against which the soft palate is elevated. From an evolutionary perspective, this ridge represents the remnant of a sphincter that encircled a more highly placed larynx; a high laryngeal position is the norm in other mammals and in the human infant (see below), but not in the human adult. Interestingly, the pharyngeal ridge becomes hypertrophic in an infant with a cleft palate, presumably in an attempt to produce a seal to the nasal airway. Ineffective velopharyngeal closure may result in nasal regurgitation of food.

The airway is sealed at the laryngeal inlet by closure of the glottis. The epiglottis is retroflexed over the laryngeal aditus as a result of passive pressure from the base of the tongue and active contraction of the aryepiglottic muscles (Ch. 35). The conventional view that laryngeal closure during swallowing occurs from inferior to superior, i.e. the vocal folds adduct first and the epiglottis covers the arytenoids and glottis last, has been challenged by studies using simultaneous electromyography and fibreoptic endoscopic evaluation of swallowing (FEES), which have reported that the aryepiglottic folds close before vocal fold adduction during a swallow. The sequence of events that close the glottis may alter according to the type of swallow and consistency of the bolus (Steele and Miller 2010). To prevent aspiration of material, irrespective of its consistency, the hyoid bone and larynx are raised and pulled anteriorly with precise timing by the suprahyoid muscles and the longitudinal muscles of the pharynx (Kendall et al 2001). In this way, the laryngeal inlet is brought forwards under the bulge of the posterior tongue, i.e. out of the path of the bolus. This action helps expand the hypopharyngeal space and relax the upper oesophageal sphincter, which is also raised by several centimetres. The bolus passes over the reflected anterior surface of the epiglottis and is swept through the laryngopharynx to the upper oesophageal sphincter.

Breathing is suspended briefly during swallowing; the larynx is closed, the soft palate is elevated and there is active neural inhibition of ventilation. Swallowing is normally initiated during an expiration, which is then inhibited while swallowing occurs (typically for a period of 0.5–1.5 seconds). The resumption of expiration provides a degree of protection against any residues of solid food that might remain uncleared within the pharynx following the swallow being inhaled into the airway. The bolus is moved by sequential contraction of the pharyngeal constrictor muscles superiorly behind the bolus. At the same time, the pharynx is shortened and elevated as palatoglossus and palatopharyngeus contract against the raised and fixed soft palate. The sequential contraction of the constrictor muscles is often assumed to be the driving force that propels the bolus towards the oesophagus. However, evidence that the head of the bolus moves faster than the wave of pharyngeal contraction suggests that, at least in some situations, the kinetic energy imparted to the bolus as it is expelled from the mouth into the oropharynx may be sufficient to carry it through the pharynx. This energy is generated by pressure gradients created within the pharynx by the tongue driving force, the hypopharyngeal suction pump, and the 'stripping action' of the pharyngeal constrictors (Nishino et al 1985).

The tongue driving force (tongue thrust pressure force) is a positive pressure that squeezes the bolus towards the laryngopharynx. It is generated by the upward movement of the tongue pressing the bolus against the contracting pharyngeal wall and requires a tight nasopharyngeal seal (created by elevation of the soft palate). There is a view that the tongue driving force is the most important factor responsible for moving the bolus down the pharynx. The hypopharyngeal suction pump is caused by the elevation and anterior movement of the hyoid and larynx, which creates a negative pressure in the laryngopharynx, drawing the bolus towards the oesophagus, aided by a more negative pressure inside the oesophagus. The pharyngeal constrictors generate a positive pressure wave behind the bolus. Their sequential contraction may facilitate clearance ('stripping') of the pharyngeal walls and piriform sinuses; if this is so, residues that remain in the valleculae must reflect inadequate tongue force generation at the end of the oral phase of swallowing.

At the end of this phase, the bolus is propelled towards the upper oesophageal sphincter. At rest, this sphincter is closed by active contraction. Cricopharyngeus relaxes prior to the bolus arriving and the sphincter is then opened actively by the combined action of the suprahyoid muscles in moving the larynx anteriorly and superiorly, and passively by pressure from the arriving bolus. The upper oesophageal sphincter thus differs in its action from that of other sphincters where opening is generally passive and a consequence of pressures generated by the movements towards them of fluids or solids.

Gag reflex

Traditionally, the stimulus for triggering a swallow has been regarded as contact with the posterior wall of the pharynx, since this is usually where the gag reflex is triggered. However, many regions of the oropharynx, when appropriately stimulated by the presence of food or liquid, are capable of triggering a swallow, although some regions are more sensitive than others, e.g. the area over the palatoglossal arches. Moreover, there appears to be little relationship between a functioning gag reflex and the ability to swallow normally. Individuals with a reduced or absent gag reflex can swallow safely; conversely, the presence of a brisk and clear gag reflex is not always associated with the ability to swallow safely.

OESOPHAGEAL PHASE

The third, or oesophageal, stage begins after the relaxation of the upper oesophageal sphincter has allowed the bolus to enter the oesophagus. This is a true peristaltic movement, in that a muscular relaxation in front of the bolus and subsequent constriction behind the bolus move it towards the stomach. Sequential waves of contractions of the oesophageal musculature now propel the bolus down to the lower oesophageal sphincter, which opens momentarily to admit the bolus to the stomach. The oesophageal phase of swallowing is much more variable than the other phases and lasts between 8 and 20 seconds.

SWALLOWING PATTERN GENERATOR

The patterning and timing of striated muscle contraction during swallowing are generated at a brainstem level in a network of neural circuits that collectively form a central swallowing pattern generator. Experimental neurophysiological and tracer studies in animals have shown that the swallowing pattern generator includes several brainstem motor nuclei and two groups of interneurones in the dorsal and ventral medulla: a dorsal swallowing group and a ventral swallowing group. The dorsal swallowing group is located in the nucleus of the solitary tract and possibly in the adjacent reticular formation, and contains neurones that generate the swallowing pattern, probably triggered by convergent information from both cortical and peripheral inputs. The afferent feedback from the branches of the superior laryngeal nerve that innervate the valleculae, epiglottis and supraglottic part of the larynx, and which relay through the nucleus of the solitary tract, facilitates laryngeal closure during swallowing (Jafari et al 2003). The ventral swallowing group is in the ventrolateral medulla above the nucleus ambiguus and contains neurones that act as 'switches', distributing the swallowing drive to motoneurone pools in the trigeminal, facial and hypoglossal nuclei and in the nucleus ambiguus.

The patterns of activation in the smooth muscle of the lower part of the oesophagus are generated locally in intramural plexuses driven by vagal autonomics.

Supramedullary influence on swallowing

Techniques including transcranial magnetic stimulation, cortical evoked potentials, positron emission tomography and functional magnetic resonance imaging (fMRI) have all been exploited in the study of volitional swallowing in humans (Harris et al 2005, Thexton 1998, Thexton and Crompton 1998). Thus far, these studies have shown that numerous cortical and subcortical regions (primary motor and sensory cortices, operculum, supplementary motor area and cingulate cortex, insula, parietal cortical areas, basal regions and cerebellum) are recruited in swallowing, although, as yet, the functional contributions made by each region have not been established, and it is not known how they influence the swallowing pattern generator. The most consistent cortical activation occurs in the primary motor and somatosensory cortices. Perhaps not surprisingly, the greatest area of fMRI activation in the primary motor cortex occurs over the portion of the precentral gyrus where the face, tongue and pharynx are represented; there is evidence for functional asymmetry between the right and left oral sensorimotor cortices (Martin et al 2004). The activity of the swallowing pattern generator is coordinated with other medullary reflexes, e.g. it is difficult to elicit a swallow when the cortical masticatory centres are stimulated. (For further reading on the motor control of oropharyngeal swallowing, see Humbert and German (2013), and on the role of sensory feedback from the oropharynx in swallowing, see Michou and Hamdy (2009).)

SWALLOWING IN THE NEONATE

In the adult, the tip of the epiglottis is significantly lower than the inferior edge of the soft palate. In the neonate, the larynx is high in the neck and the epiglottis may extend above the soft palate so that the laryngeal airway is in direct continuity with the posterior nares (**Figs 34.11–34.12**); a potential space is therefore formed between the soft palate above, the epiglottis behind and the tongue anteroinferiorly. In other mammals with an oropharyngeal anatomy similar to that of the human infant, up to 14 cycles of tongue movement or oral phases cause the accumulation of food in this space. Subsequent emptying of the space is a single event followed by movement of the bolus down the oesophagus. The ratio of accumulation cycles to swallow events in the human neonate is approximately 1.5 : 1, which is lower than in other mammals but still implies some temporary accumulation. In the case of a liquid bolus, accumulated material may be passed laterally to the epiglottis through the piriform fossae rather than over the flexed epiglottis, although it is not known whether this happens in the human infant. (For further reading, see Delaney and Arvedson (2008).)

Swallow safety is critical at all ages. In the neonate, where coordination of suckling, swallowing and breathing is not fully developed, the potential risk of airway obstruction and/or aspiration of ingested milk or other material during swallowing is reduced because the intranarial larynx prevents the bolus from entering the larynx before and after a

Dysphagia is a symptom of many diseases and can arise from anatomical or functional deficits anywhere along the path from the oral cavity to the stomach or from a failure of neural control resulting from congenital malformations, trauma, stroke or neurodegenerative diseases within the central nervous system. Loss of muscle strength affecting tongue propulsion or pharyngeal contraction will impair the initiation of the swallow and/or bolus transport. This will result in retention of residues within the pharynx following a swallow. Residues may also accumulate in the valleculae and piriform sinuses. Failure to protect the airways adequately may result in the penetration of swallowed material food or liquid into the vestibule of the larynx. Aspiration will result if the material then passes through the vocal folds, which may result in airway obstruction or aspiration pneumonia. Penetration may sometimes occur in normal individuals.

Loss of control of the pharyngeal phase of swallowing, e.g. due to neurological disease or ablative head and neck surgery, may result in aspiration of food, especially fluids, leading to pneumonia. This problem may be addressed surgically by pharyngostomy or epiglottopexy. In cervical pharyngostomy, a tube may be passed through the cervical skin, fascia and platysma directly into the piriform fossa (Patil et al 2006). This is achieved by passing a curved forcep into the piriform fossa and pushing it laterally, displacing the contents of the carotid sheath and tenting up the platysma and cervical skin from the inside. By cutting down on to the forcep, it is possible to grasp the feeding tube and pull it into the piriform fossa prior to feeding it on into the oesophagus. The tract formed by such a puncture epithelializes and may be used for long-term alimentation. In epiglottopexy, the neck and the pharynx are opened to expose the laryngeal inlet, the aryepiglottic folds are denuded of mucosa to encourage their adhesion, and the epiglottis is sutured down to the aryepiglottic folds to shield the laryngeal inlet. The resultant compromise of the airway can be offset by the creation of an alternative airway via a tracheostomy. The pharyngeal wall and cervical skin are reconstituted by suturing.

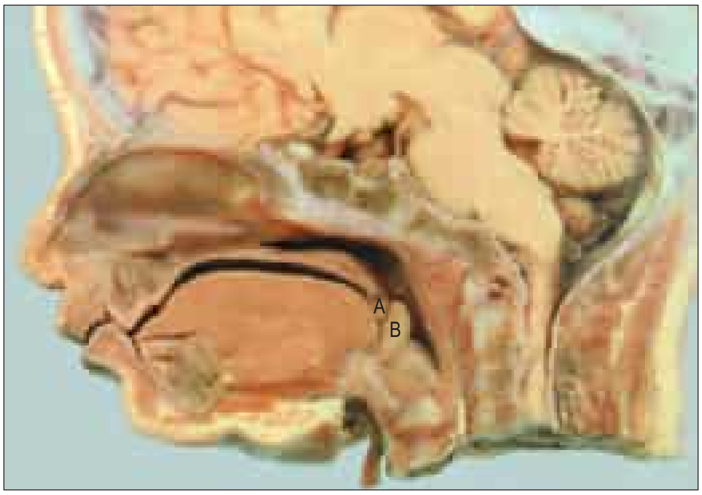

Fig. 34.11 A sagittal section of the head of a neonate. Note the relatively high position of the larynx, the opening being at the level of the soft palate (A). Other abbreviations: B, epiglottis. (With permission from Berkovitz BKB, Holland GR, Moxham BJ 2002 Oral Anatomy, Embryology and Histology, 3rd edn. Edinburgh: Mosby.)

swallow. (Neonates born at term demonstrate an increase in suck and swallow rates over the course of the first 4 weeks; efficiency of feeding, measured as volume of nutrient per suck and per swallow, doubles over the course of the neonatal period (Qureshi et al 2002).)

The change towards adult anatomy and coordination of the phases of swallowing starts a few months after birth (Kelly et al 2007). Differential growth in length of the human pharynx causes the larynx to take up its low adult position and the epiglottis to lose contact with the soft palate; the larynx reaches its final position around the time of puberty. The adult anatomy does not allow any significant accumulation of food to occur immediately anterior to the epiglottis, which means that the transport of food through the fauces has to bear a 1 : 1 relationship to pharyngeal and oesophageal transit. Moreover, the lowered larynx compromises the previously protected airway; the hyoid and larynx are therefore raised and pulled forwards during a swallow in order to minimize the risk of deglutitive aspiration.

DYSPHAGIA

Available with the *Gray's Anatomy* e-book

PHARYNGOSTOMY AND EPIGLOTTOPEXY

Available with the *Gray's Anatomy* e-book

Bonus e-book image

Fig. 34.5 Thornwaldt's cyst.

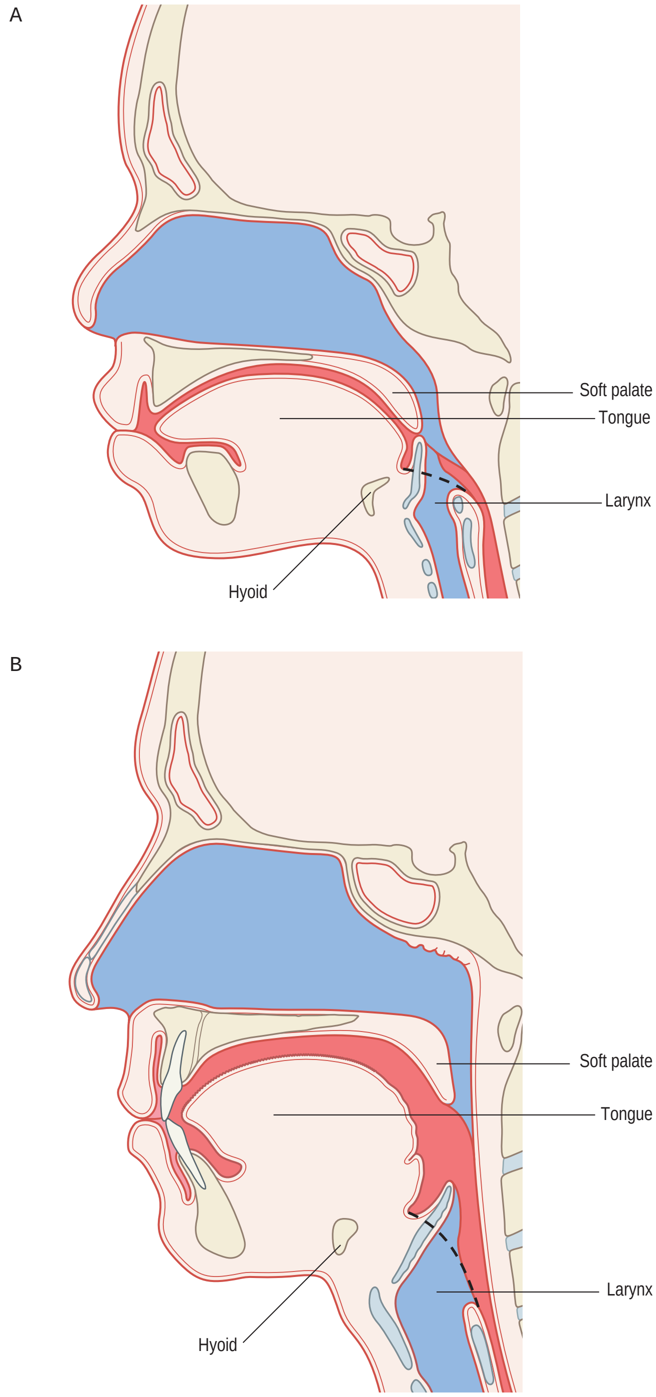

Fig. 34.12 A sagittal section of the head and neck in an infant (**A**) and adult (**B**) human. The food way and the airway are shaded in red and blue, respectively. **A**, In the infant human, the oral cavity is small and the tongue and palate are flatter. **B**, In the adult human, the larynx is lower in the neck, and the food way and airway cross in the pharynx. (Redrawn with permission from Matsuo K, Palmer JB 2008 Anatomy and physiology of feeding and swallowing – normal and abnormal. Phys Med Rehabil Clin N Am 19:691–707.)

KEY REFERENCES

Flint PW, Haughey BH, Lund VJ et al 2010 Cummings Otolaryngology: Head and Neck Surgery, vols 2 and 3, 5th ed. Philadelphia: Elsevier, Mosby.
Two volumes that contain comprehensive descriptions of the anatomy, physiology and pathology of the pharynx and palate.

Graney DO, Retruzzelli GJ, Myers EW 1998 Anatomy. In: Cummings CW, Fredrickson JM, Harker LA et al (eds) Otolaryngology: Head and Neck Surgery, vol 2, 3rd ed. St Louis: Elsevier, Mosby; pp. 1327–48.
A concise account of the anatomy of the pharynx, highlighting features of clinical relevance.

Hollinshead WH 1982 Anatomy for Surgeons, vol I: The Head and Neck, 3rd ed. Philadelphia: Harper & Row.
An older textbook that provides a valuable account of the anatomy of the pharynx and of tissue spaces in the neck. It is also a thorough guide to the earlier literature.

Matsuo K, Palmer JB 2008 Anatomy and physiology of feeding and swallowing – normal and abnormal. Phys Med Rehabil Clin N Am 19: 691–707.
A review of swallowing focussing on the process model of feeding.

O'Reilly RC, Sando I 2010 Anatomy and physiology of the Eustachian tube. In: Flint PW, Haughey BH, Lund VJ et al (eds) Cummings Otolaryngology: Head and Neck Surgery, 5th ed. St Louis: Elsevier, Mosby; Ch. 131, pp. 1866–75.
A review of the clinical anatomy and physiology of the pharyngotympanic tube.

REFERENCES

Arens R, McDonough JM, Corbin AM et al 2002 Linear dimensions of the upper airway structure during development: assessment by magnetic resonance imaging. Am J Respir Crit Care Med 165:117–22.

Bluestone CD 1998 Anatomy and physiology of the Eustachian tube. In: Cummings CW, Fredrickson JM, Harker LA et al (eds) Otolaryngology: Head and Neck Surgery, vol 4, 3rd ed. St Louis: Elsevier, Mosby; pp. 3003–25.

Bonnington A, Mahon M, Whitmore I 1988 A histological study of the cricopharyngeus muscle in man. J Anat 156:27–37.

Brownlow H, Whitmore I, Willan PLT 1989 A quantitative study of the histochemical and morphometric characteristics of the human cricopharyngeus muscle. J Anat 166:67–75.

Cho JH, Kim JK, Lee H-Y et al 2013 Surgical anatomy of the soft palate. Laryngoscope 123:2900–4.

Chong VFH, Ong CK 2008 Nasopharyngeal carcinoma. Eur J Radiol 66: 437–447.

Flint PW, Haughey BH, Lund VJ et al 2010 Cummings Otolaryngology: Head and Neck Surgery, vols 2 and 3, 5th ed. Philadelphia: Elsevier, Mosby.
Two volumes that contain comprehensive descriptions of the anatomy, physiology and pathology of the pharynx and palate.

Dallan I, Seccia V, Muscatello L et al 2011 Transoral endoscopic anatomy of the parapharyngeal space: a step-by-step logical approach with surgical considerations. Head Neck 33:557–61.

Delaney AL, Arvedson JC 2008 Development of swallowing and feeding: prenatal through first year of life. Dev Disabil Res Rev 14:105–17.

Fogel RB, Malhotra A, White DP 2004 Sleep 2: Pathophysiology of obstructive sleep apnoea/hypnoea syndrome. Thorax 59:159–63.

Freelander E 1992 Blood supply of the human levator and tensor veli palatini muscles. Clin Anat 5:34–44.

Graney DO, Retruzzelli GJ, Myers EW 1998 Anatomy. In: Cummings CW, Fredrickson JM, Harker LA et al (eds) Otolaryngology: Head and Neck Surgery, vol 2, 3rd ed. St Louis: Elsevier, Mosby; pp. 1327–48.
A concise account of the anatomy of the pharynx, highlighting features of clinical relevance.

Gray SD, Pinborough-Zimmerman J 1998 Velopharyngeal incompetence. In: Cummings CW et al (eds) Otolaryngology: Head and Neck Surgery, vol 5: Pediatric Otolaryngology, 3rd ed. St Louis: Elsevier, Mosby; pp. 174–87.

Harris ML, Julyan P, Kulkarni B et al 2005 Mapping metabolic brain activation during human volitional swallowing: a positron emission tomography study using [18F]fluorodeoxyglucose. J Cereb Blood Flow Metab 25:520–6.

Hiiemae KM, Palmer JB 1999 Food transport and bolus formation during complete feeding sequences on foods of different initial consistency. Dysphagia 14:31–42.

Hiiemae KM, Palmer JB 2003 Tongue movements in feeding and speech. Crit Rev Oral Biol Med 14:413–29.

Hollinshead WH 1982 Anatomy for Surgeons, vol I: The Head and Neck, 3rd ed. Philadelphia: Harper & Row.
An older textbook that provides a valuable account of the anatomy of the pharynx and of tissue spaces in the neck. It is also a thorough guide to the earlier literature.

Humbert IA, German RZ 2013 New directions for understanding neural control in swallowing: the potential and promise of motor learning. Dysphagia 28:1–10.

Ison S, Remmers JE, Tanaka A et al 1997 Anatomy of pharynx in patients with obstructive sleep apnea and in normal subjects. J Appl Physiol 84:1319–26.

Jafari S, Prince RA, Kim DY et al 2003 Sensory regulation of swallowing and airway protection: a role for the internal superior laryngeal nerve in humans. J Physiol 550:287–304.

Johal AMA, Patel SI, Battagel JM 2007 The relationship between craniofacial anatomy and obstructive sleep apnoea: a case-controlled study. J Sleep Res 16:319–26.

Kelly BN, Huckabee ML, Jones RD et al 2007 The first year of human life: coordinating respiration and nutritive swallowing. Dysphagia 22: 37–43.

Kendall KA, Leonard RJ, McKenzie SW 2001 Accommodation to changes in bolus viscosity in normal deglutition: a videofluoroscopic study. Ann Otol Rhinol Laryngol 110:1059–65.

Lang IM, Shaker R 2000 An overview of the upper esophageal sphincter. Curr Gastroenterol Rep 2:185–90.

Malhotra A, White DP 2002 Obstructive sleep apnoea. Lancet 360: 237–45.

Maran AGD, Mackenzie IJ, Murray JAM 1984 The parapharyngeal space. J Laryngol Otol 98:371–89.

Martin RE, MacIntosh BJ, Smith RC et al 2004 Cerebral areas processing swallowing and tongue movement are overlapping but distinct: a functional magnetic resonance imaging study. J Neurophysiol 92:2428–43.

Matsuo K, Palmer JB 2008 Anatomy and physiology of feeding and swallowing – normal and abnormal. Phys Med Rehabil Clin N Am 19: 691–707.
A review of swallowing focussing on the process model of feeding.

Michou E, Hamdy S 2009 Cortical input in control of swallowing. Curr Opin Otolaryngol Head Neck Surg 17:166–71.

Mioche L, Hiiemae KM, Palmer JB 2002 A postero-anterior videofluorographic study of the intra-oral management of food in man. Arch Oral Biol 47:267–80.

Moore EJ, Janus J, Kasperbauer J 2012 Transoral robotic surgery of the oropharynx: clinical and anatomic considerations. Clin Anat 25: 135–41.

Nishino T, Yonezawa T, Honda Y 1985 Coordination of swallowing and respiration in unconscious subjects. J Appl Physiol 70:988–93.

O'Reilly RC, Sando I 2010 Anatomy and physiology of the Eustachian tube. In: Flint PW, Haughey BH, Lund VJ et al (eds) Cummings Otolaryngology: Head and Neck Surgery, 5th ed. St Louis: Elsevier, Mosby; Ch. 131, pp. 1866–75.
A review of the clinical anatomy and physiology of the pharyngotympanic tube.

Orita Y, Sando I, Hirsch BE et al 2002 Postnatal development of the eustachian tube glands. Laryngoscope 112:1647–52.

Palmer JB, Hiiemae KM, Matsuo K et al 2007 Volitional control of food transport and bolus formation during feeding. Physiol Behav 91: 66–70.

Papaioannou G, Kambas I, Tsaoussoglou M et al 2013 Age-dependent changes in the size of adenotonsillar tissue in childhood: implications for sleep-disordered breathing. J Pediatr 162:269–74.

Patil PM, Warad NM, Patil RN et al 2006 Cervical pharyngostomy: an alternative approach to enteral feeding. Oral Surg Oral Med Oral Pathol Oral Radiol Endod 102:736–40.

Qureshi MA, Vice FL, Taciak VL et al 2002 Changes in rhythmic suckle feeding patterns in term infants in the first month of life. Dev Med Child Neurol 44:34–9.

Sade J (ed) 1989 Basic Aspects of the Eustachian Tube and Middle Ear Disease. Geneva: Kugler & Ghedini.

Sakamoto Y 2013 Interrelationships between the innervations from the laryngeal nerves and the pharyngeal plexus to the inferior pharyngeal constrictor. Surg Radiol Anat 35:721–8.

Sakamoto Y 2014 Gross anatomical observations of attachments of the middle pharyngeal constrictor. Clin Anat 27:603–9.

Sato K, Nakashima T 2006 Human adult deglutition during sleep. Ann Otol Rhinol Laryngol 115:334–9.

Sato K, Nakashima T 2007 Sleep-related deglutition in children. Ann Otol Rhinol Laryngol 116:747–53.

Steele CM, Miller AJ 2010 Sensory input pathways and mechanisms in swallowing: a review. Dysphagia 25:323–33.

Thexton A 1998 Some aspects of swallowing. In: Harris M, Edgar M, Meghji S (eds) Clinical Oral Science. Oxford: Wright; pp. 150–66.

Thexton AJ, Crompton AW 1998 The control of swallowing. In: Linden RWA (ed) The Scientific Basis of Eating. Taste and Smell, Salivation, Mastication and Swallowing and their Dysfunctions. Frontiers of Oral Biology Series, vol 9. Basel: Karger; pp. 168–222.

Wang C, Kundaria S, Fernandez-Miranda J et al 2014 A description of arterial variants in the transoral approach to the parapharyngeal space. Clin Anat 27:1016–22.

Wood-Jones I 1940 The nature of the soft palate. J Anat 77:147–70.

Yuca K, Varsak YK 2012 Thornwaldt's cyst. Eur J Gen Med 9:(Suppl 1) 26–9.